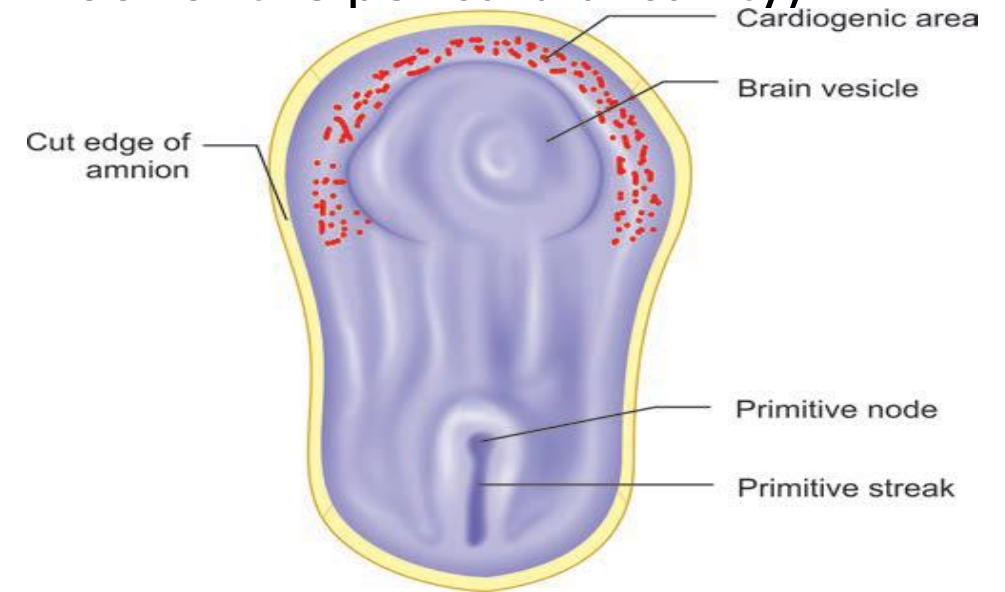
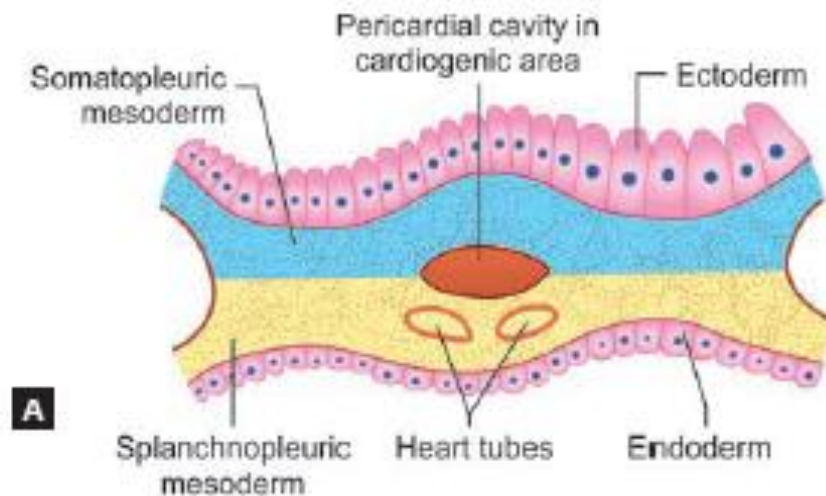


Development of the Heart

D.Ndhlovu-Chikwanda

- Heart is mesodermal in origin
- first organ of the body to start functioning
- Formed from splanchnopleuric mesoderm lying cranial to the prochordal plate
- This mesoderm constitutes the cardiogenic area
- ***primitive heart tubes*** develop in the cardiogenic area
- pericardial cavity is formed from the cranial, midline, part of the intraembryonic coelom
- With the formation of the coelom, the intraembryonic mesoderm of the region splits into :-
- ❖ a **somatopleuric layer** adjoining the ectoderm (in **roof of pericardial cavity**)
- ❖ a **splanchnopleuric layer** adjoining the endoderm (in floor of the pericardial cavity)



- **Cardiac Progenitor Cells and Primary and Secondary Heart Fields**
- *Cardiac progenitor cells* appear at the caudal epiblast, lateral to primitive streak 16th–18th days of development
- migrate cranially through the primitive streak into the splanchnopleuric layer of intraembryonic mesoderm
- form a horseshoeshaped ***primary heart field*** along the cranial end of embryonic disc rostral to buccopharyngeal membrane and neural folds
- By 21st day, a ***secondary heart field*** appears in the splanchnopleuric mesoderm ventral to posterior pharynx after the cranial folding

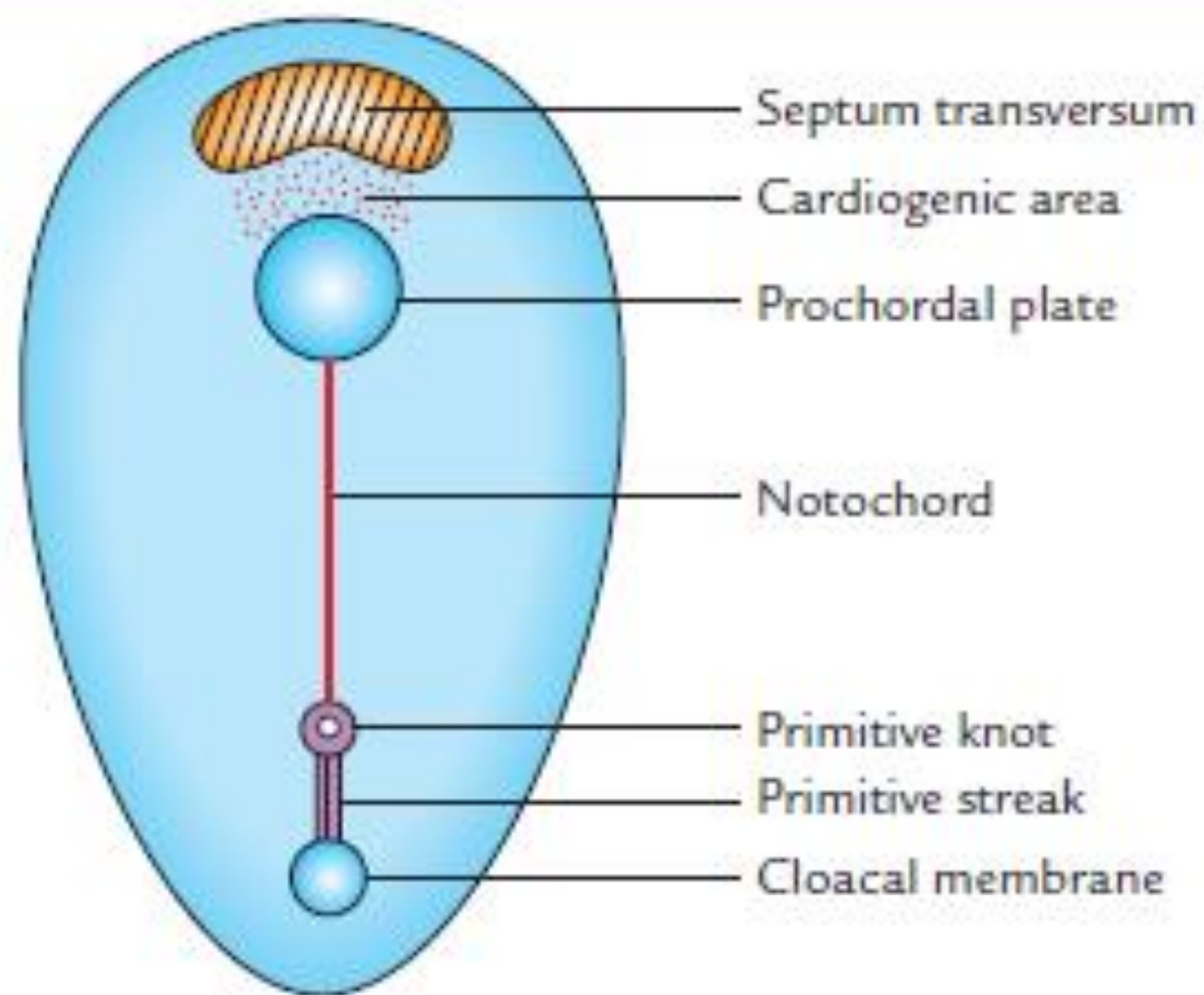


Fig. 18.1 Location of cardiogenic area.

- **Cardiogenic Area**
- Pharyngeal **endoderm** underlying primary heart field induces formation of cardiogenic area where horseshoeshaped blood islands are formed by angiogenic clusters
- Which later get canalized to form endothelial lined heart tubes that are surrounded by **myoblasts**
- area is between the dorsal wall of yolk sac & floor (splanchnopleuric
- layer) of pericardial cavity (which is derived from part of the intraembryonic coelom)

- After formation of the head fold, tube lies dorsal to the pericardial cavity & ventral to the foregut
- tube invaginates into the pericardial sac on its dorsal side
- splanchnopleuric mesoderm lining the dorsal side of the pericardial cavity proliferates to form a thick layer- the ***myoepicardial mantle*** (or *epimyocardial mantle*)
- When invagination is complete, myoepicardial mantle completely surrounds the heart tube

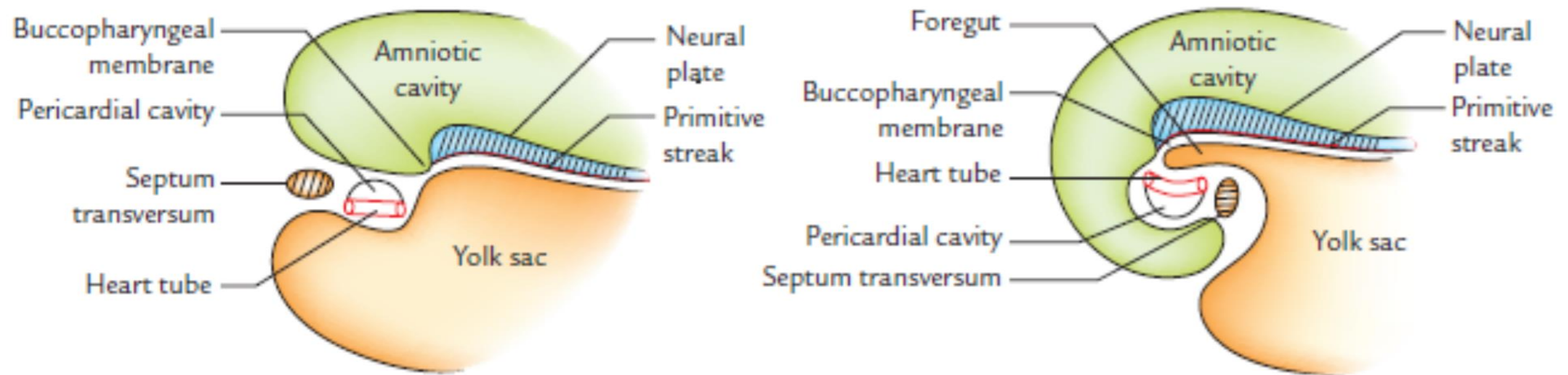


Fig. 18.5 Changing positions of the pericardial cavity and heart tube following the formation of head fold.

- Myoepicardial mantle gives rise to the cardiac muscle (myocardium) & visceral layer of pericardium (epicardium)
- parietal layer of pericardium is derived from somatopleuric mesoderm
- Endothelial heart tube forms the endocardial of the heart

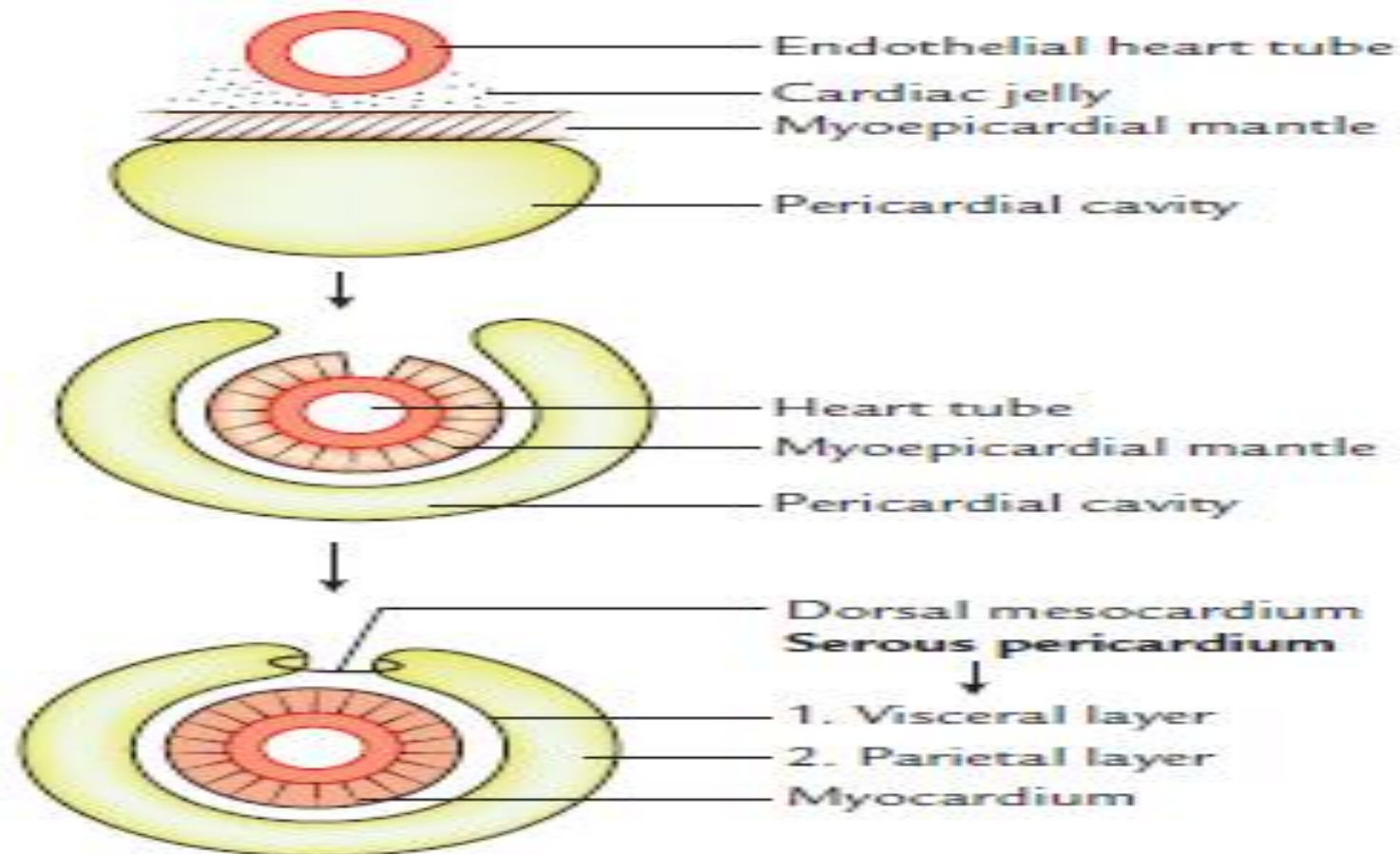
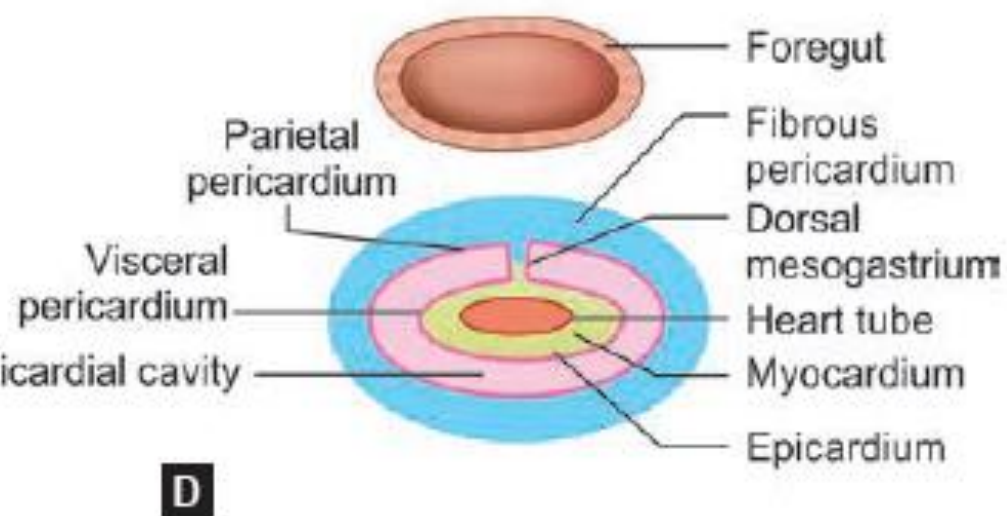
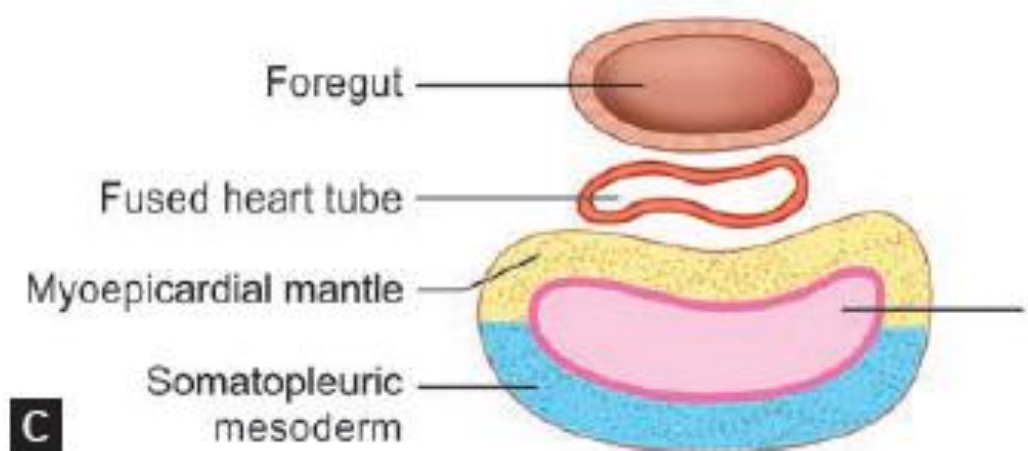
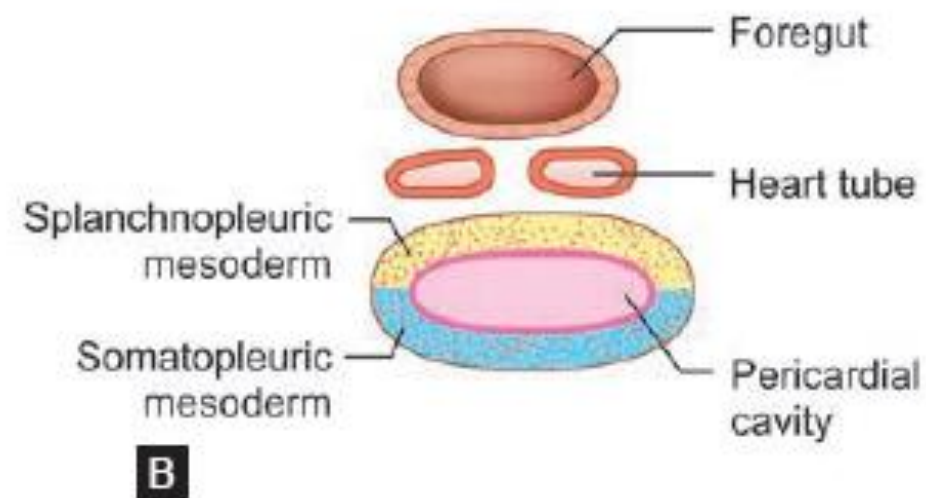
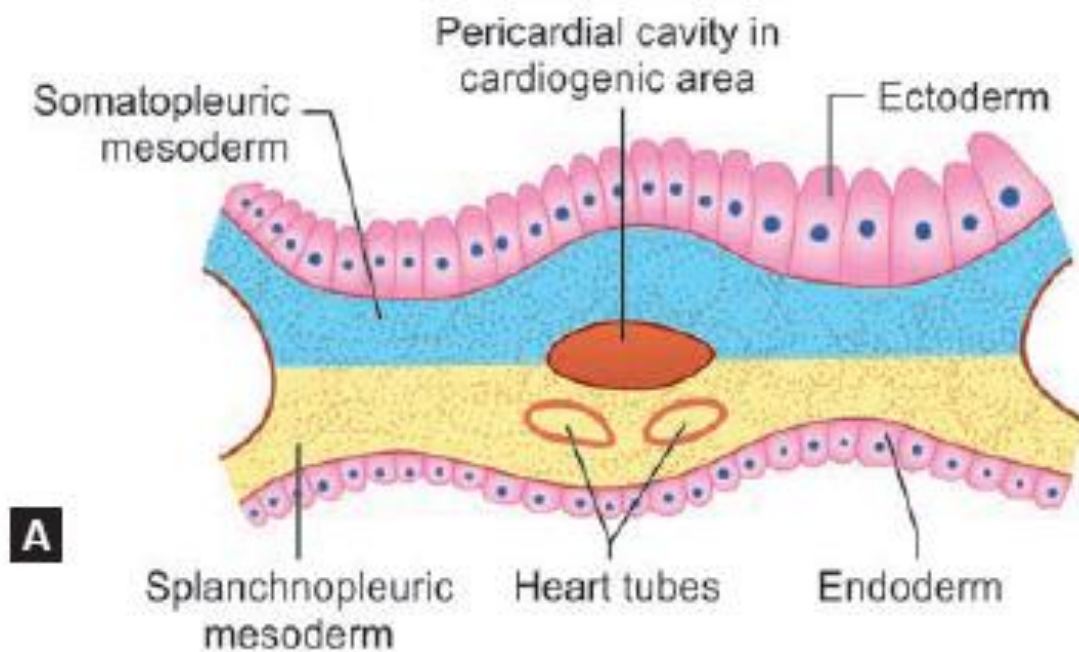
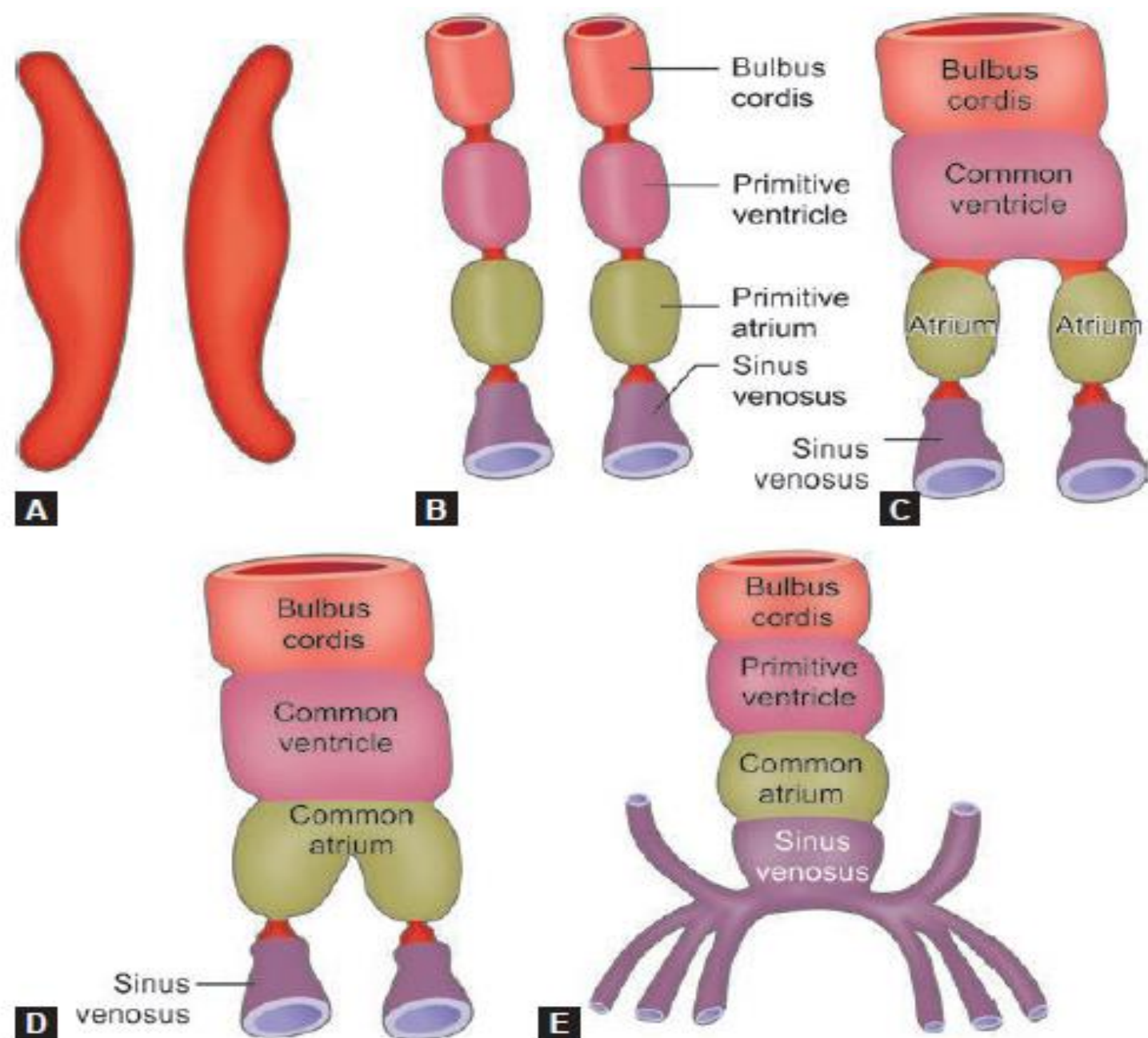


Fig. 18.6 Schematic diagram to show the development of cardiac wall.



- heart is at first seen in the form of right & left endothelial heart tubes that soon fuse with each other
- single tube thus formed shows a series of dilatations
- These are:
 - ❖ **Bulbus cordis**
 - ❖ **Ventricle** (primitive ventricle)
 - ❖ **Atrium** (primitive atrium or atrial chamber)
 - ❖ **Sinus venosus.**
- sinus venosus & atrium are connected by ***sinoatrial orifice***.
- atrium & ventricle are in communication through ***atrioventricular canal (AV canal)***.



Figs 15.7A to E: (A) Right and left heart tubes; (B to E) Progressive fusion of tubes from cranial

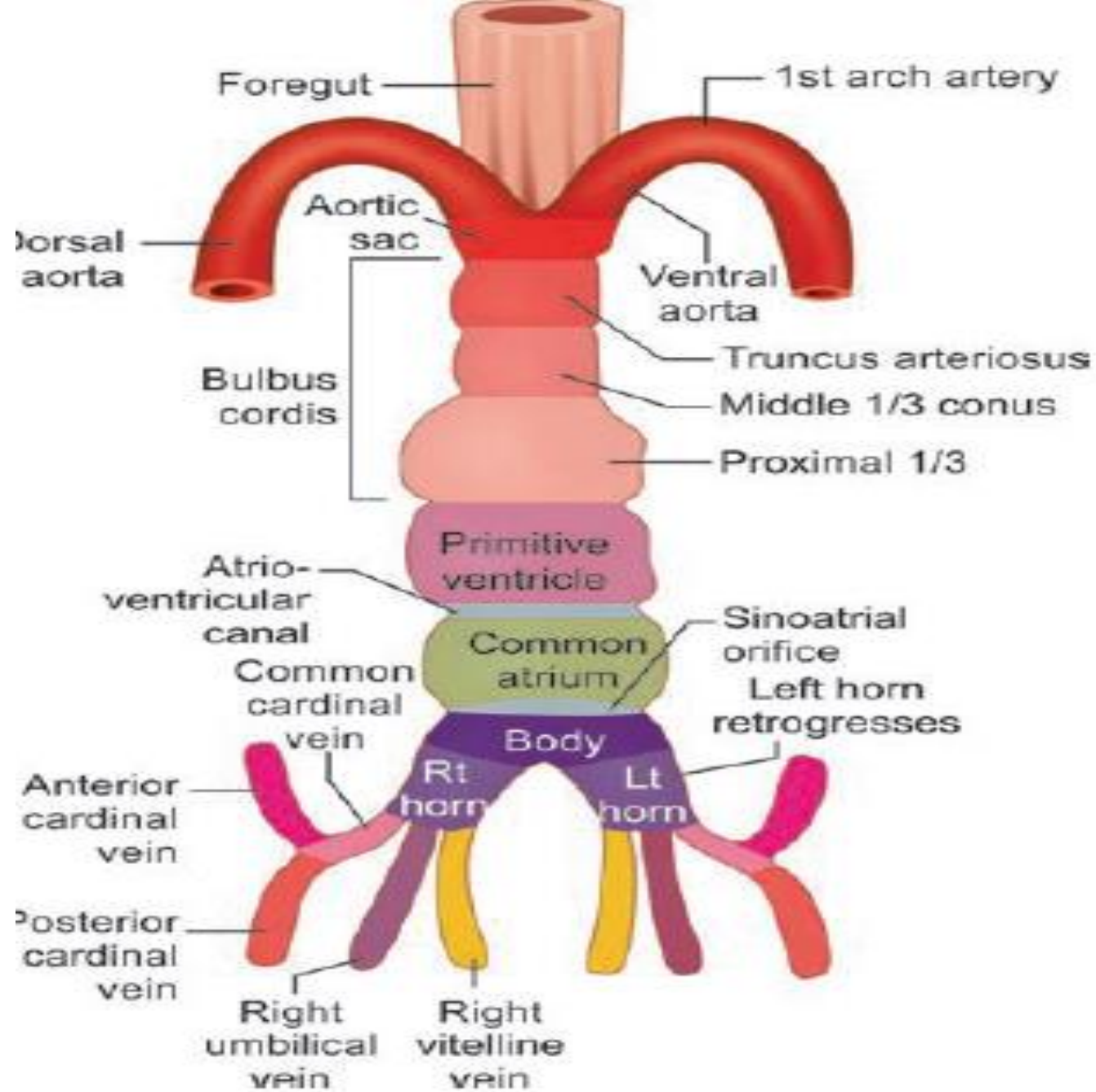


Fig. 15.8: Subdivisions of fused heart tube

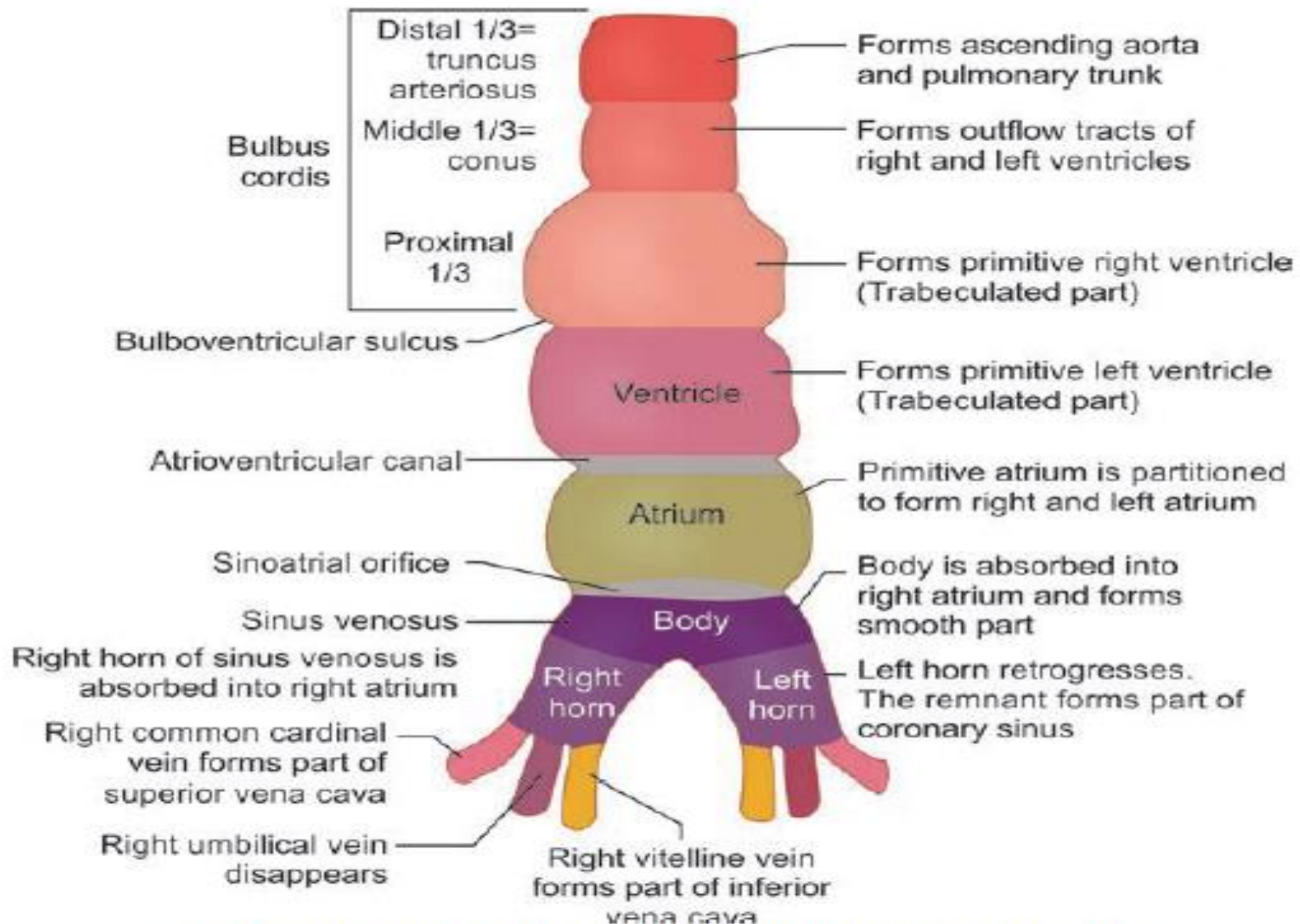
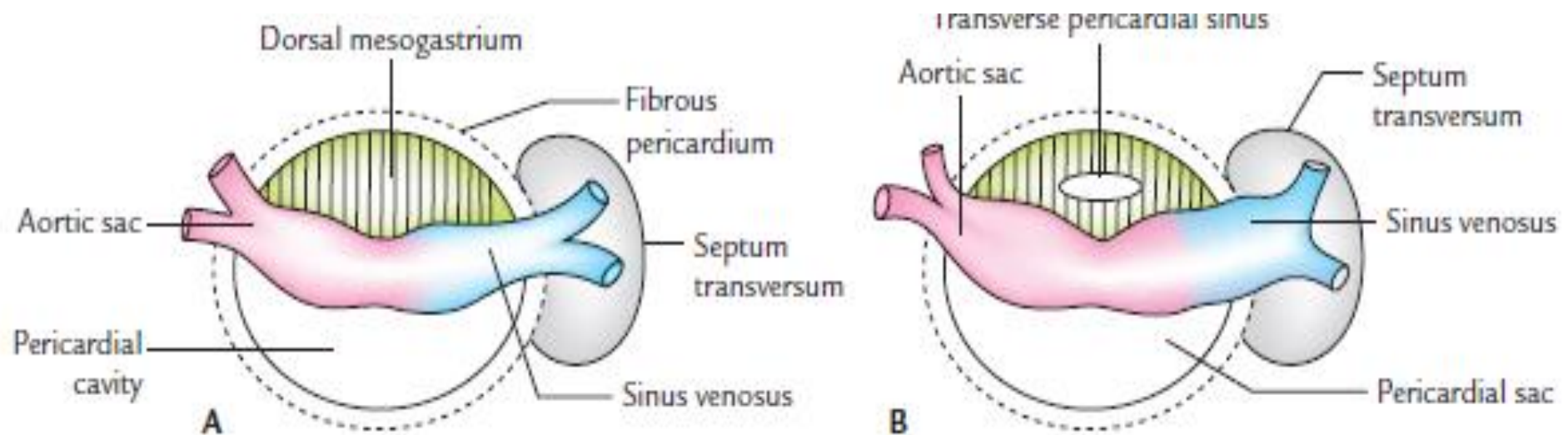


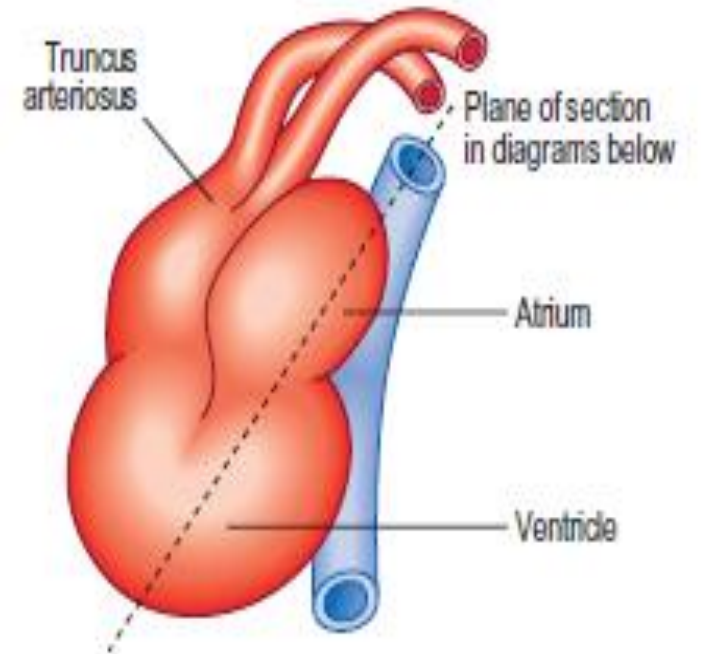
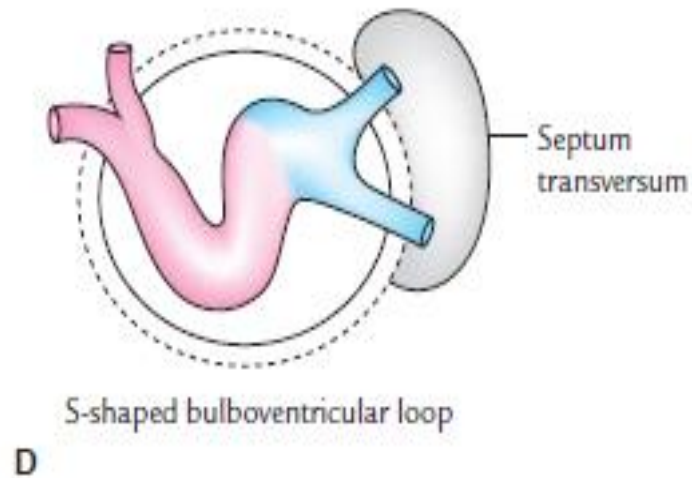
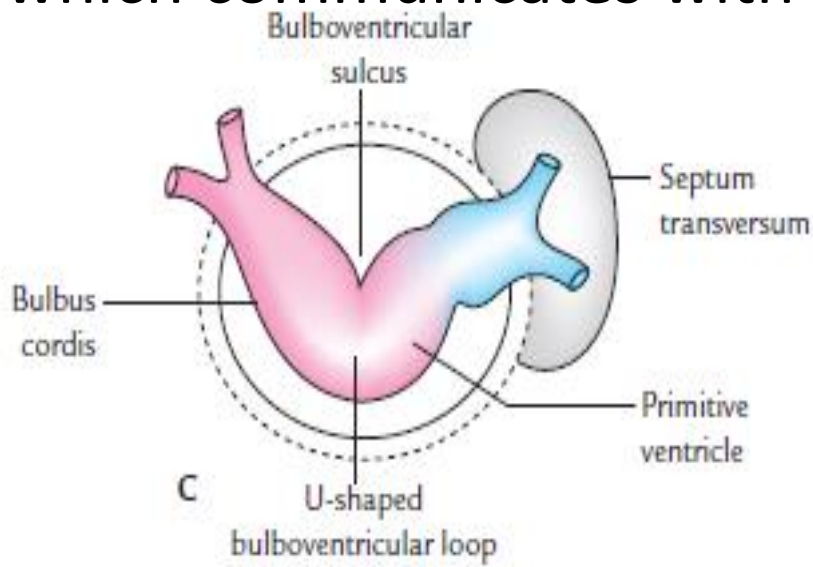
Fig. 15.9: Main subdivisions of the heart tube and their fate

• EXTERIOR OF THE HEART

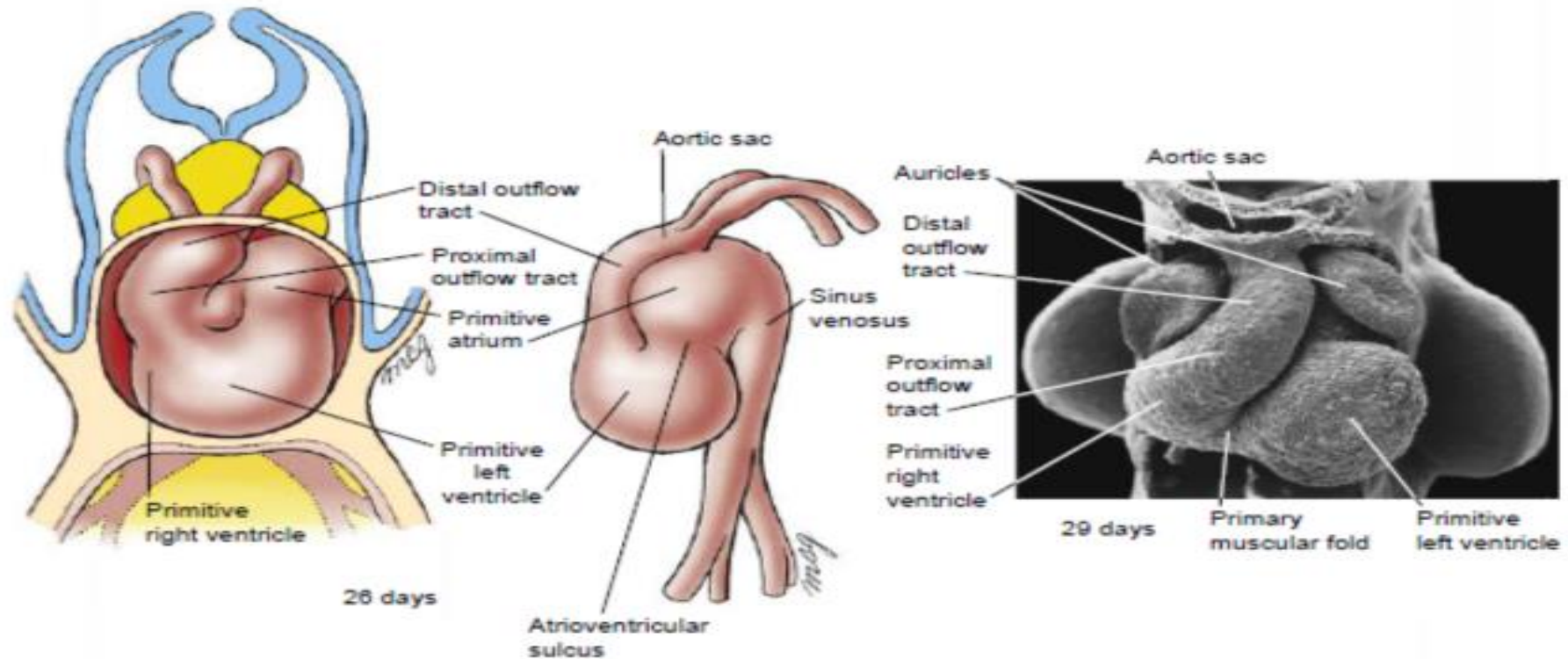
- heart tube is placed longitudinally
- suspended from the dorsal wall of the pericardial cavity by the ***dorsal mesocardium***
- mesocardium disappears & heart tube lies free within the pericardial sac, suspended by its 2 ends
- caudal part of the heart tube (atrium, sinus venosus) is embedded within the substance of the septum transversum
- heart tube lying within the pericardial cavity is bulbus cordis and ventricle



- Bulbus cordis & ventricle grows rapidly
- folds on itself to form a “U”-shaped ***bulboventricular loop***
- primitive atrium is to the left & dorsal to primitive ventricle
- atrium & sinus venosus are freed from the septum transversum, come to lie behind & above the ventricle, heart tube is now “S”-shaped
- bulbus cordis & ventricle, are separated by ***bulboventricular sulcus***
- sulcus becomes shallower & conus, proximal bulbus cordis & ventricle, come to form one chamber
- which communicates with the truncus arteriosus



- atrial chamber lies behind the upper part of the ventricle & of the truncus arteriosus expands
- sinus venosus moves away from septum transversum & occupies a position dorsal to primitive atrium
- As a result of these changes, the exterior of the heart assumes its definitive shape



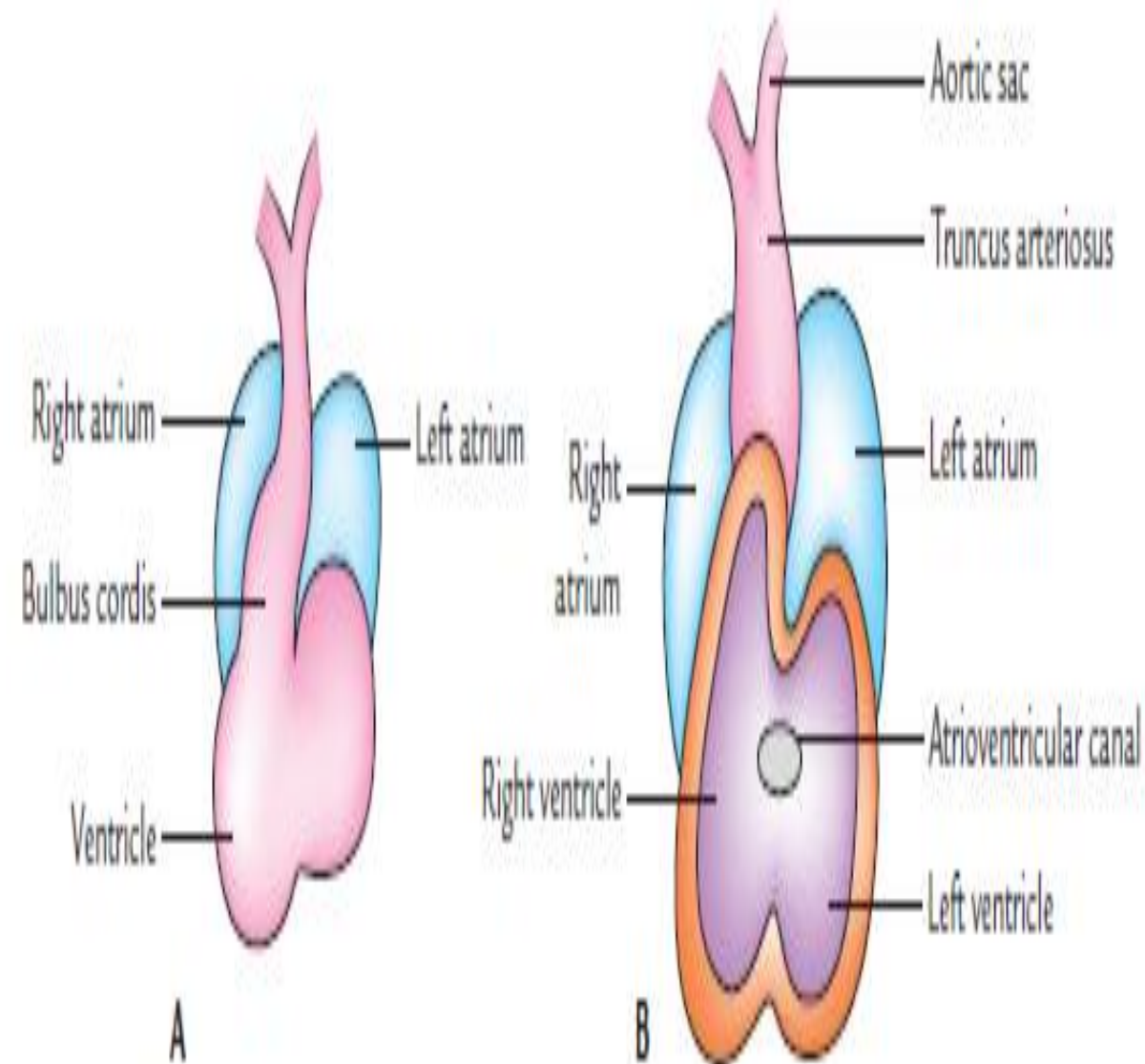
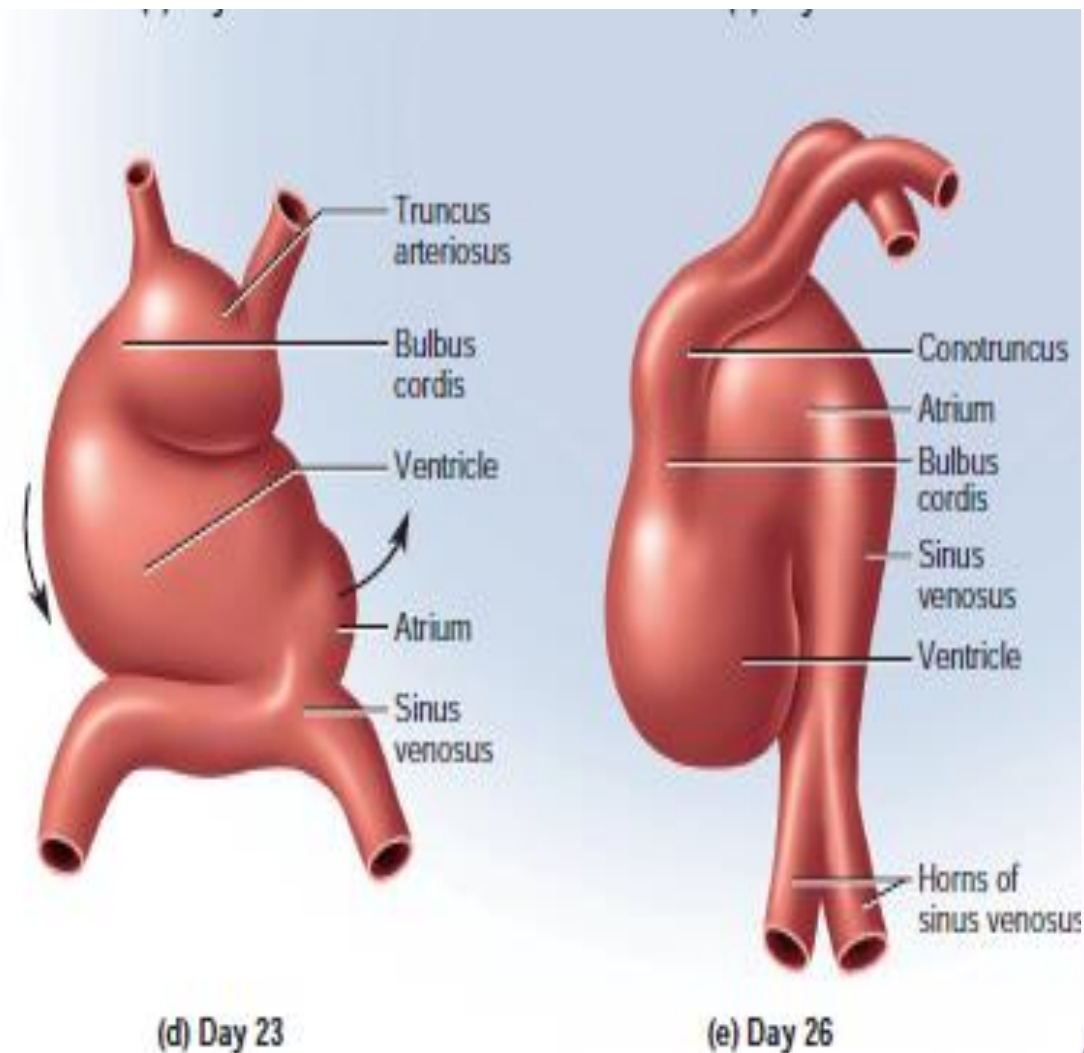


Fig. 18.8 Establishment of the external form of the heart (A). The interiors of developing ventricles are shown on the right (B).

- Cardiac septations to form definitive heart occur between 5th & 8th weeks

- **DEVELOPMENT OF VARIOUS CHAMBERS OF THE HEART**

- **1. Development of Atria**

- *Sinus Venosus and Its Absorption into the Right Atrium*

- caudal (venous end) part of primitive heart tube

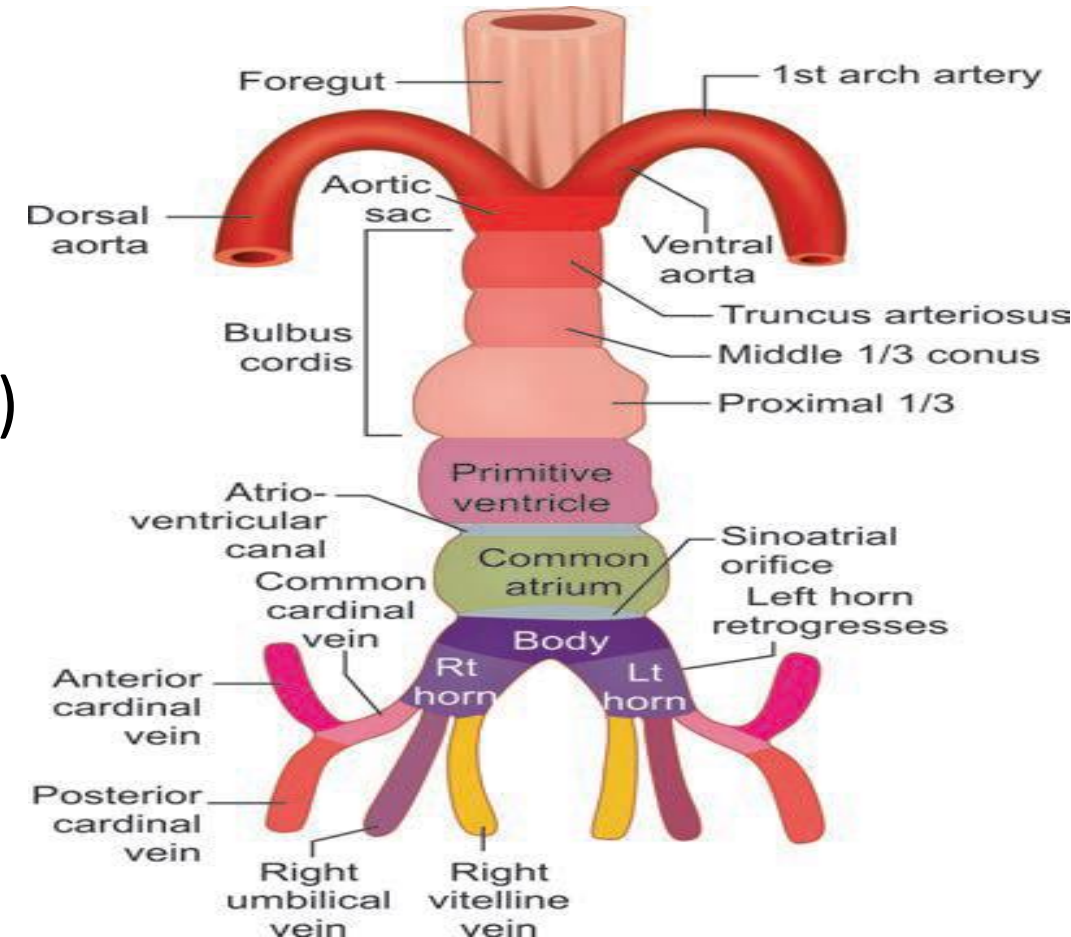
- presents a body with ***right and left horns***

- ❖ One *vitelline vein* (from the yolk sac),

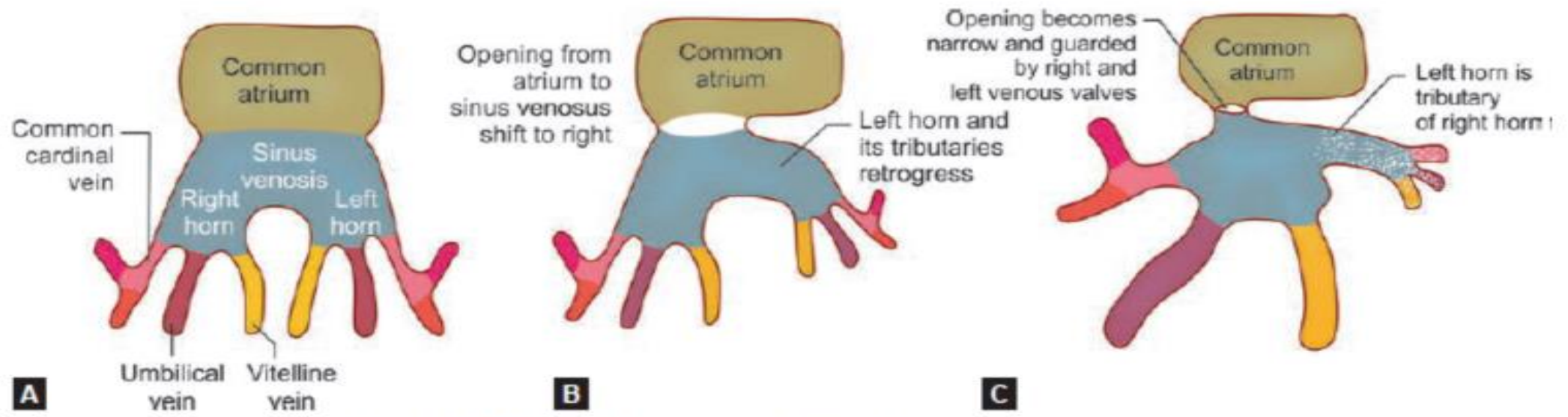
- ❖ one *umbilical vein* (from the placenta)

- ❖ one *common cardinal vein* (from the body wall)

- Join each horn of the sinus venosus

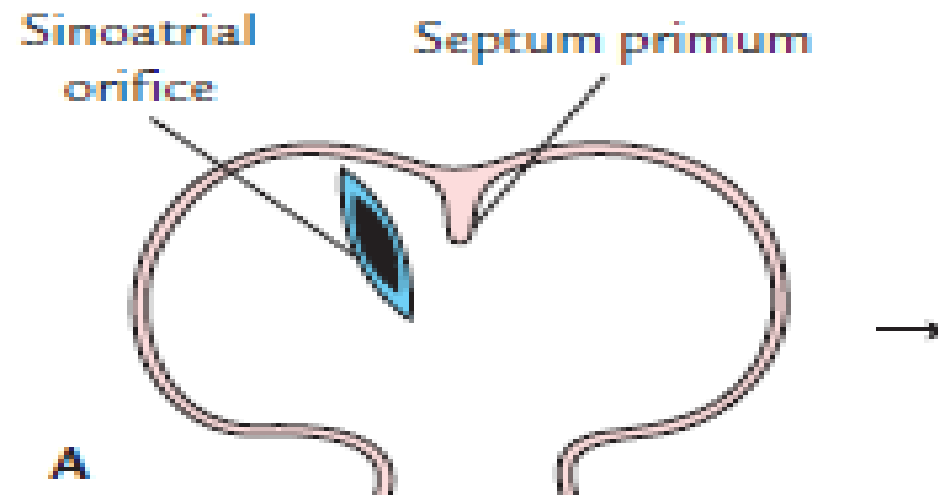


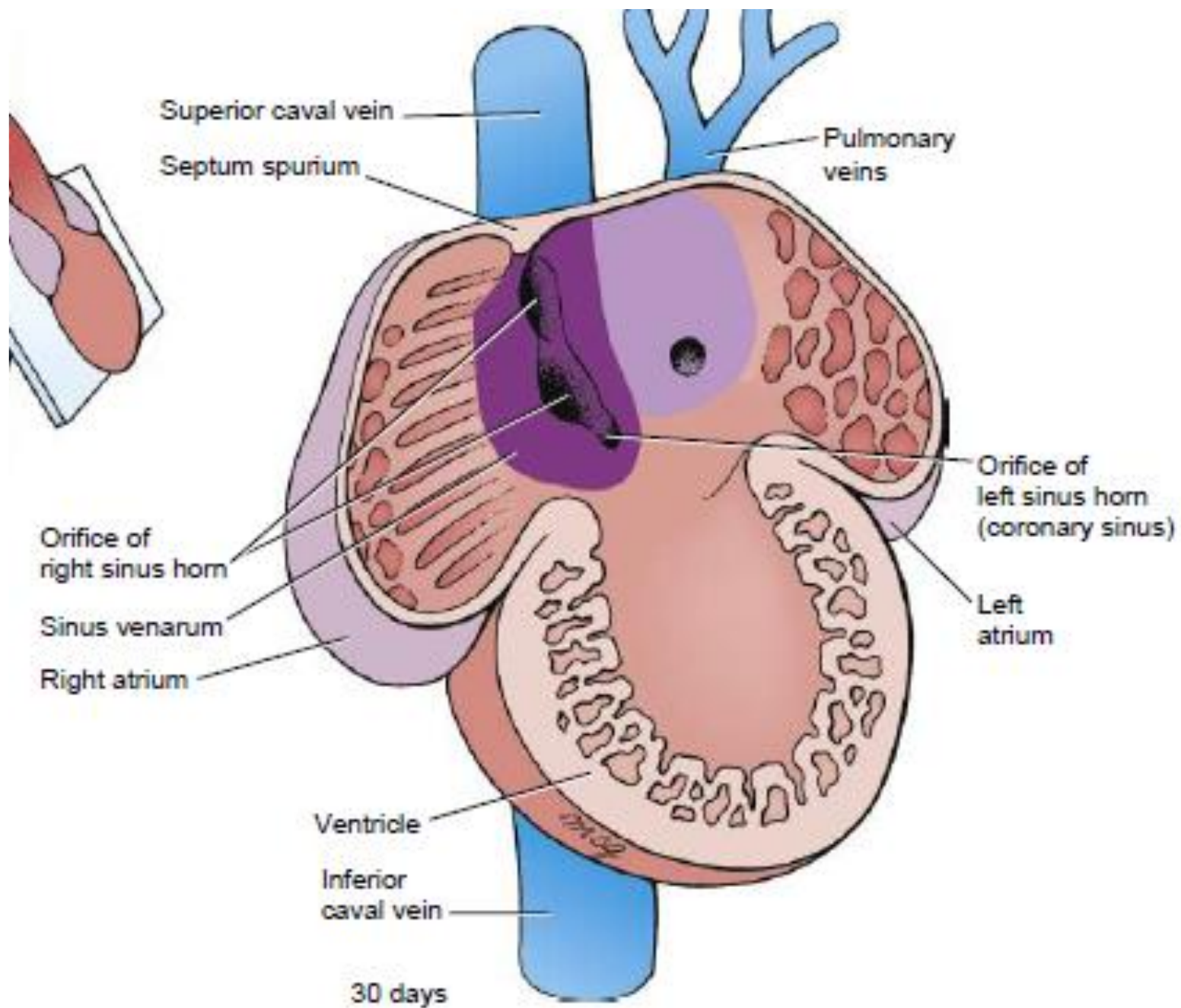
- sinus venosus & primitive atrial chamber connected by an opening ***sinoatrial orifice***
- partially separated by *grooves* that appear on the lateral wall, at the junction of these 2 chambers
- *right groove* remains shallow but the *left* deepens
- results the left part of the sinus venosus becomes separated from the atrial chamber.



Figs 15.11A to C: Retrogression of the left horn of sinus venosus

- blood enters the atrium through the right half of the sinus
- left horn of the sinus venosus & its tributaries reduces in size
- left horn becomes part of the ***coronary sinus***
- ***Change in size and orientation of sinoatrial orifice***
- Initially sinoatrial orifice is larger in size, transverse in orientation & median in position
- opening narrows & shifts to the right & the dorsal aspect of primitive atrium
- changes from transverse to oval & finally to vertical with a narrow slit





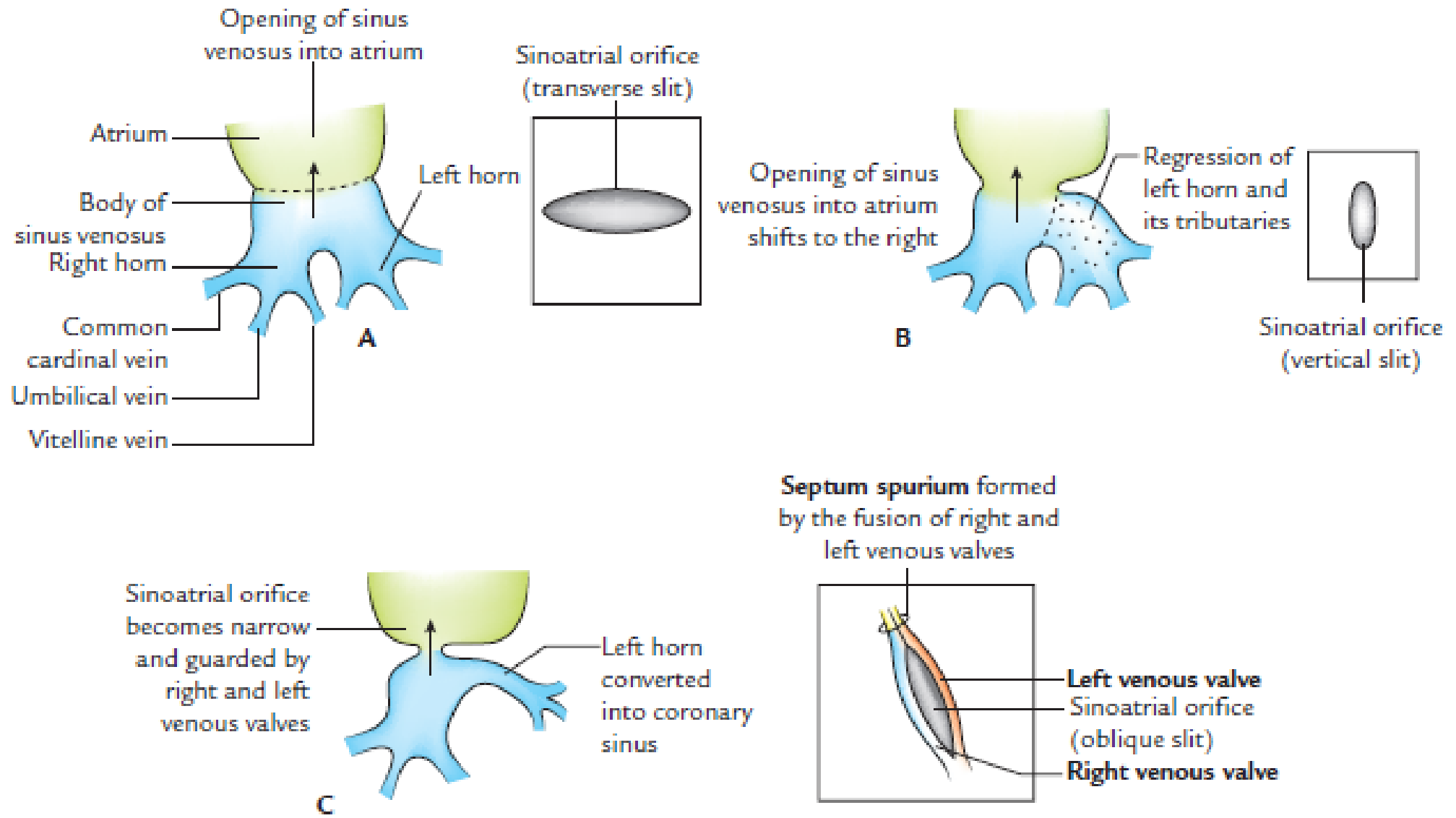
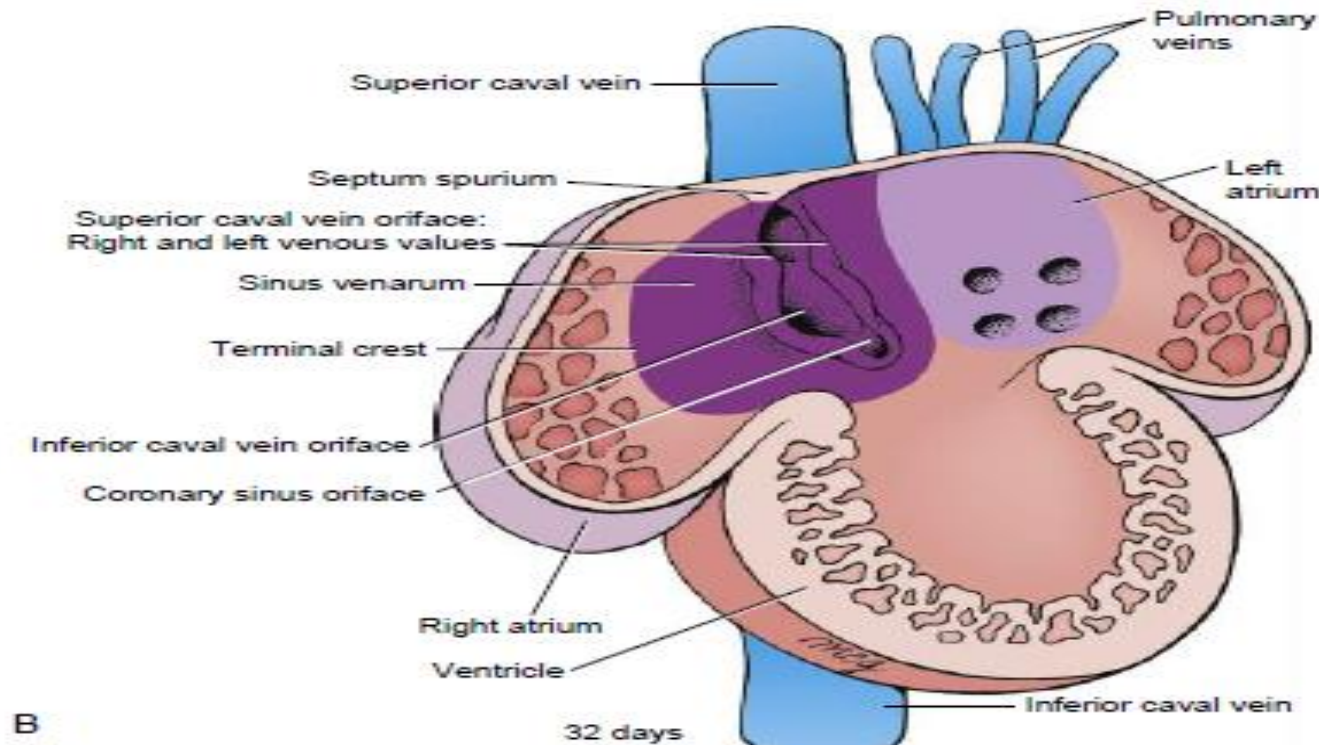


Fig. 18.13 Regression of sinus venosus. The changes in shape of sinusoidal orifice are shown in the inset on the right side: A. transverse slit, B. vertical slit, and C. oblique slit.

- Cranially the 2 valves fuse to form ***septum spurium*** Caudally forms ***sinus septum***
- ***Fate of tributaries of sinus venosus:***
- Each common cardinal vein receives the venous blood from cranial and caudal parts of embryo via anterior and posterior cardinal veins respectively
- **right common cardinal vein** becomes part of the ***superior vena cava***
- **right vitelline vein** forms the **terminal part of the *inferior vena cava***



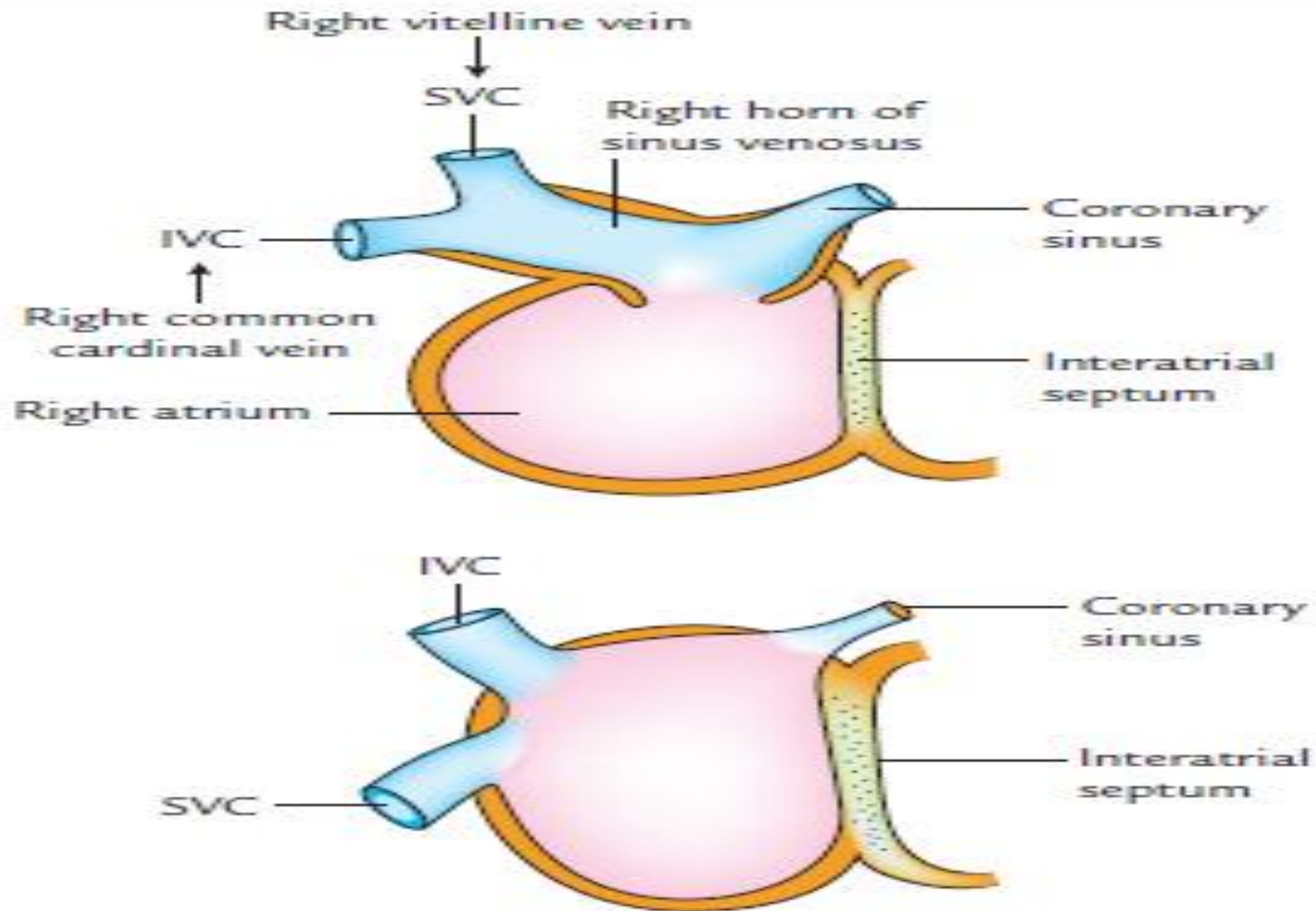
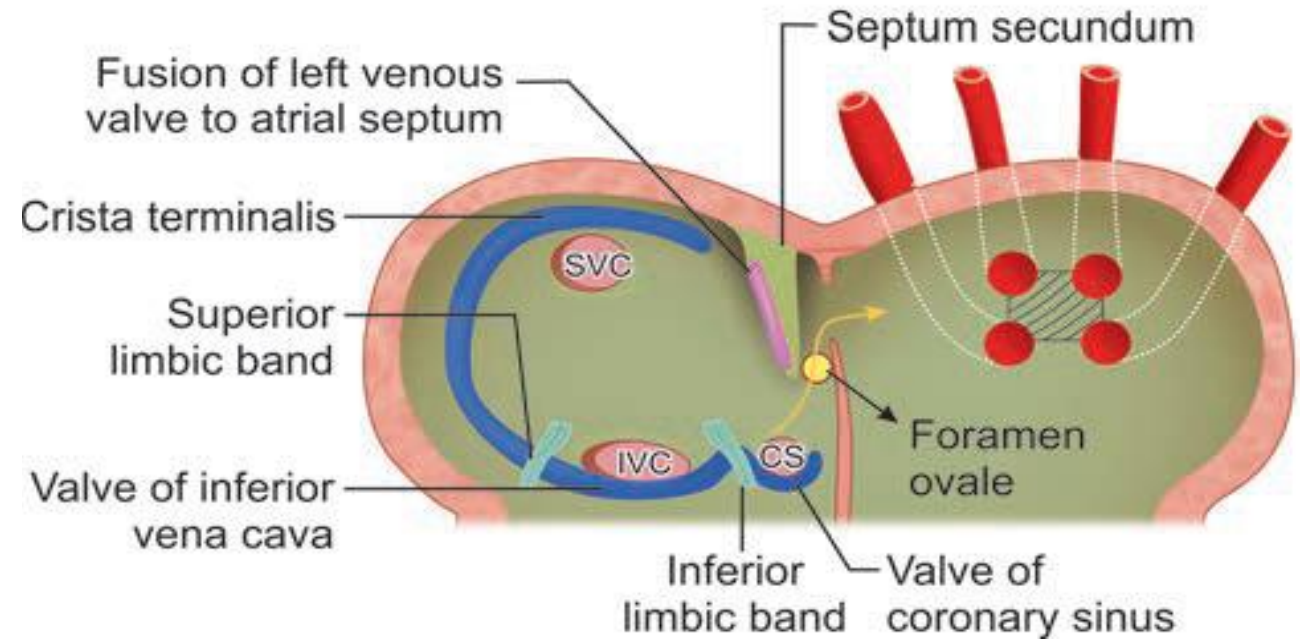
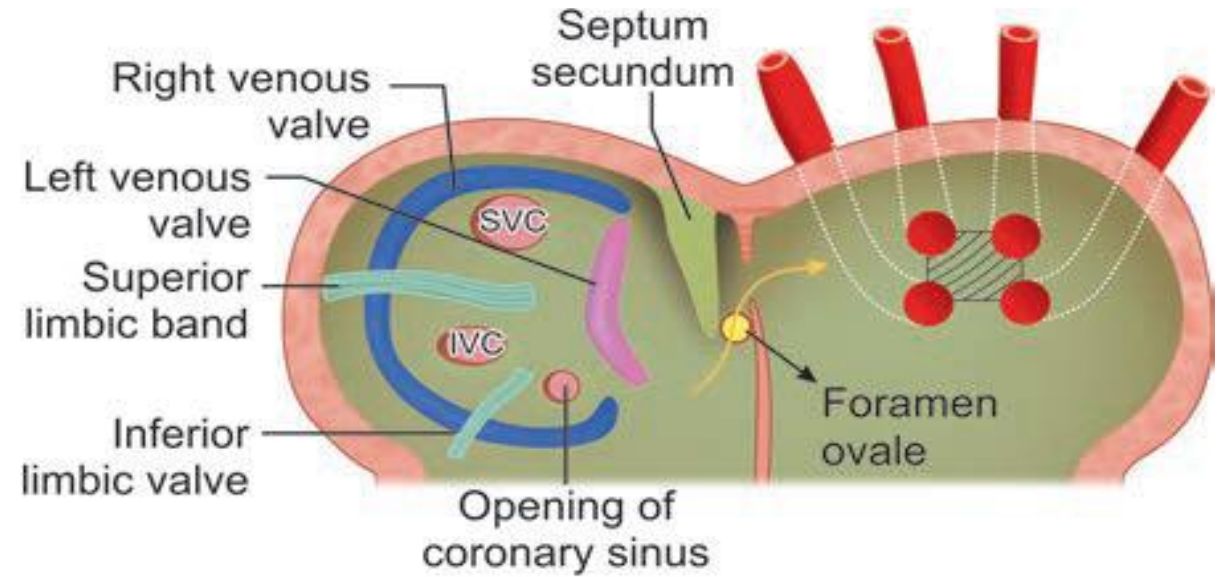


Fig. 18.14 Absorption of sinus venosus into the right atrium. SVC = superior vena cava; IVC = inferior vena cava.

- ***Fate of right and left venous valves:***
- right margin of the original sinoatrial orifice expands greatly
- divides into 3 parts by 2 muscular bands
- ***superior & inferior limbic bands***
- 3 parts of right venous valve are the
- 1 ***crista terminalis***
- 2 ***valve of the inferior vena cava***
- 3 ***valve of the coronary sinus***



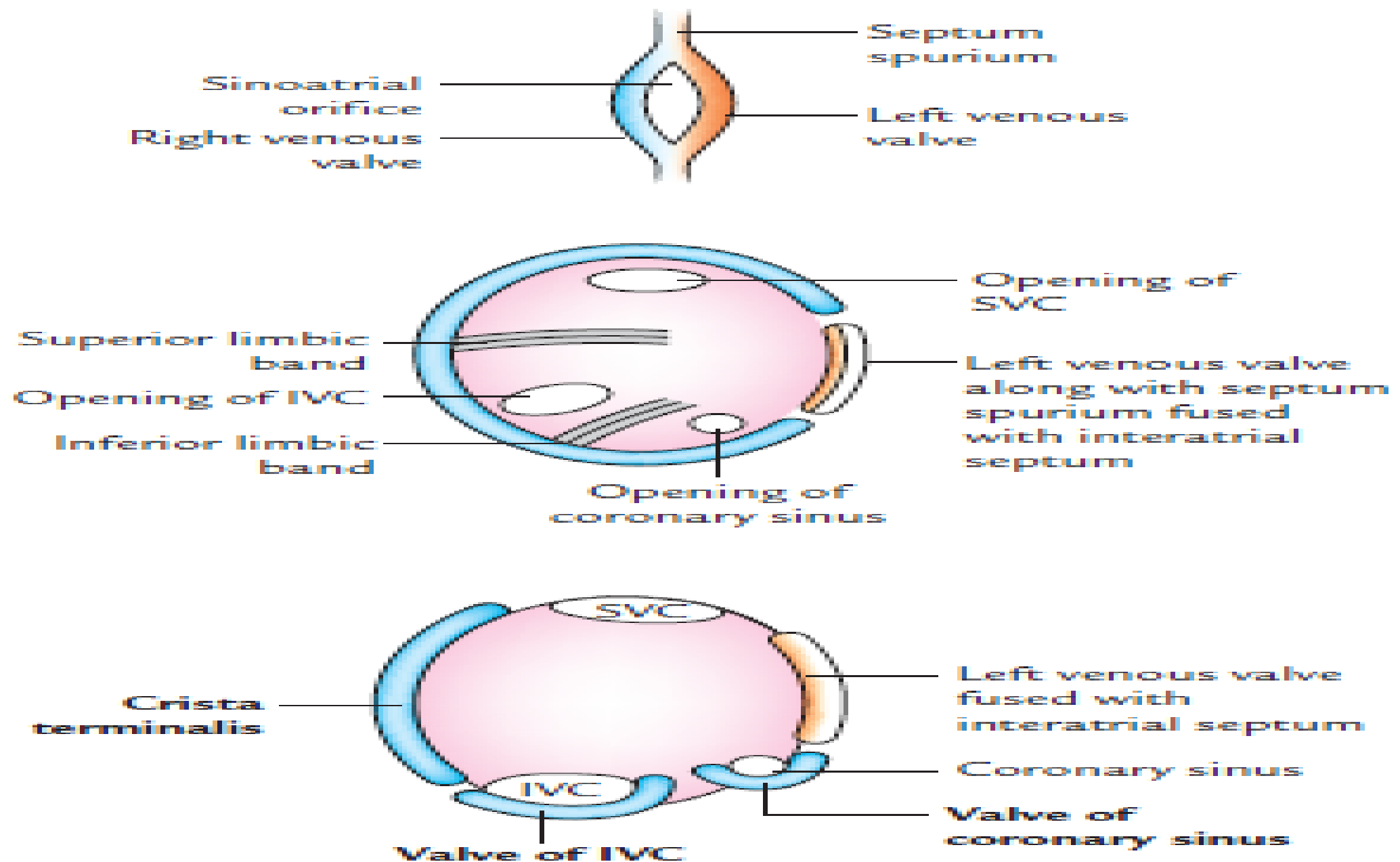
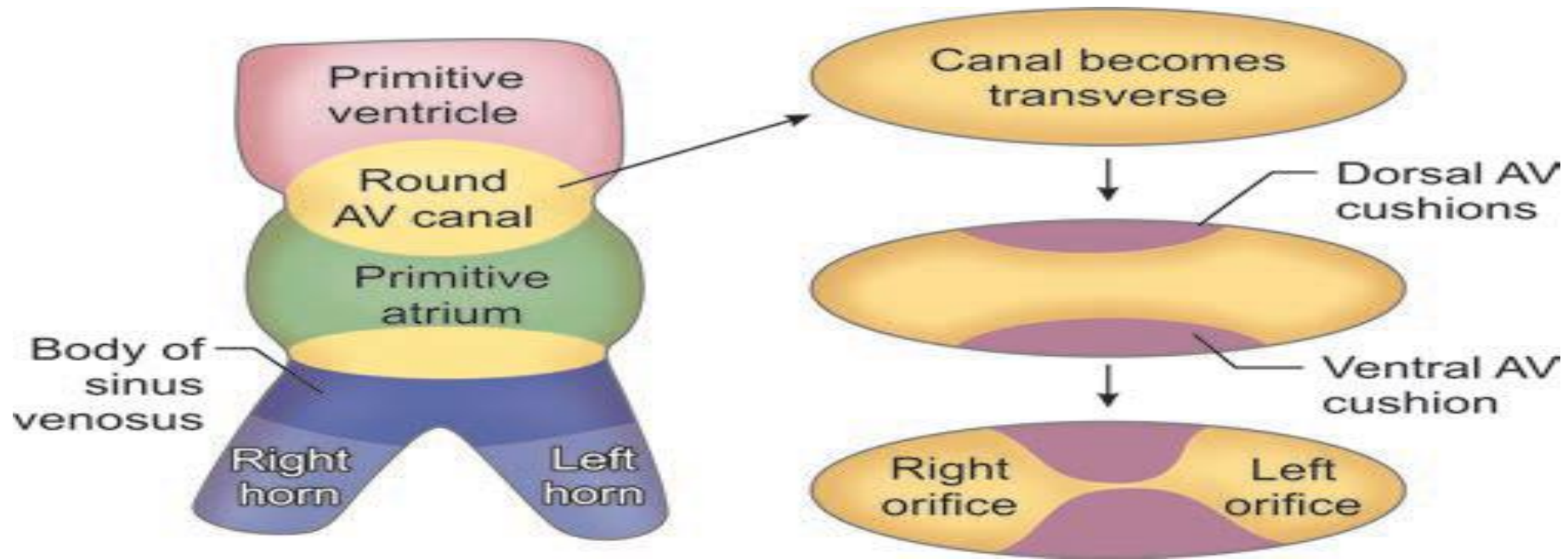


Fig. 18.15 Fate of right and left venous valves.

- **Atrioventricular Canal**
- communication between common atrial chamber & ventricle
- AV canal divides into right & left halves
- circular in shape initially and becomes transverse
- 2 thickenings, the AV endocardial cushions appear on the dorsal and ventral walls of AV canal by proliferation of subendocardial mesenchymal cells around right and left AV canals
- grow toward each other and fuse form the ***septum intermedium***



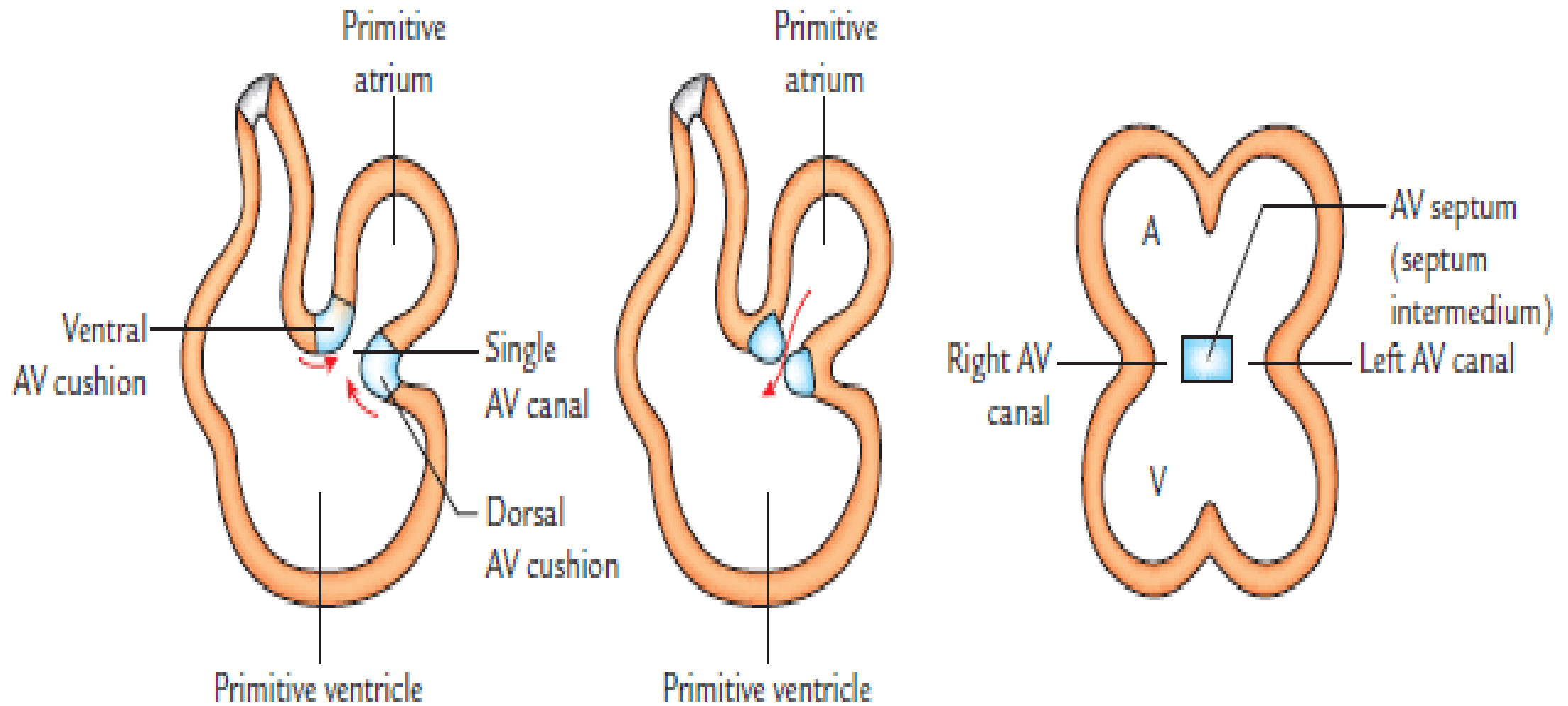
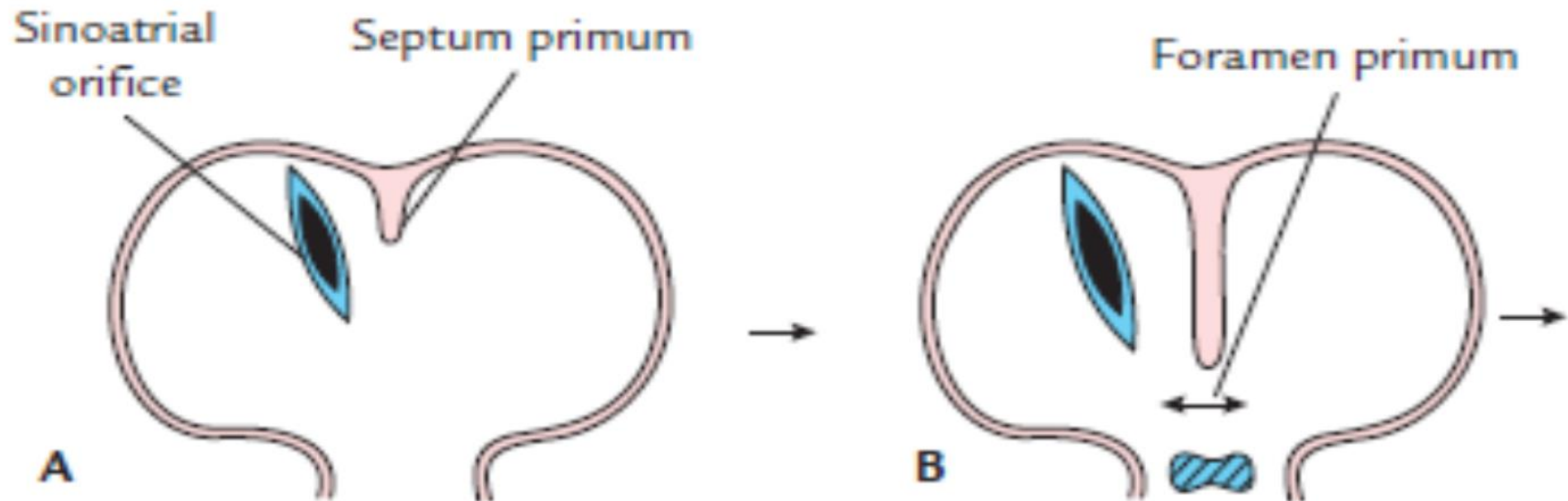


Fig. 18.11 Development of atrioventricular (AV) septum.

- ***Formation of Interatrial Septum***
- atrial chamber undergoes division into right & left halves by formation of 2 septa
- sickle-shaped fold from the roof & dorsal wall of primitive atrium is formed
- Known as *septum primum* located in the midline & to the left of septum spurium
- grows downward toward AV canal & septum intermedium.
- ultimately fuses with the septum intermedium



- Before septum primum fuses with the septum intermedium, blood flows through the gap between them - ***foramen primum/ ostium primum***
- Before foramen primum can be closed, another path for flow of blood is created
- created by breaking down of the upper part of the septum primum
- new gap is - ***foramen secundum***
- septum primum has a free upper edge

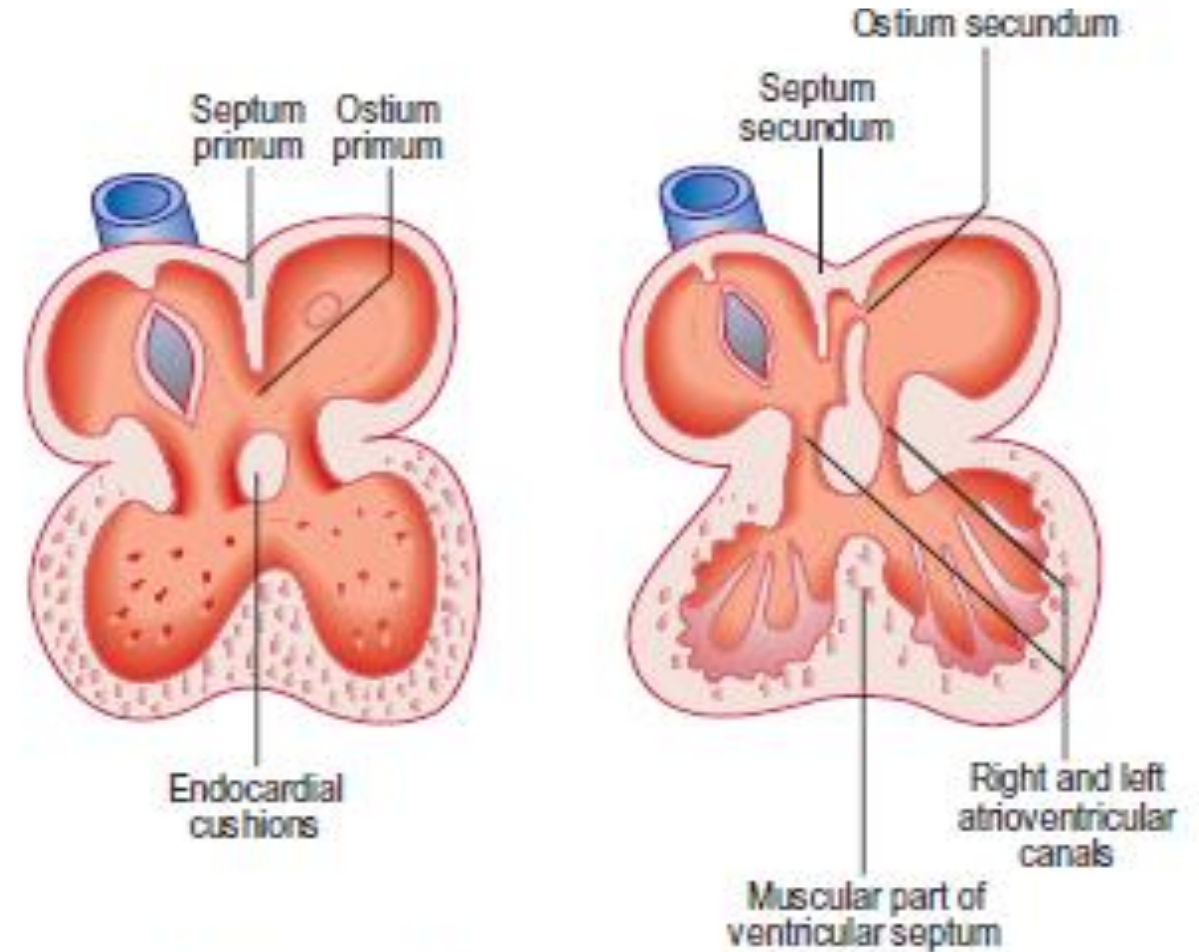
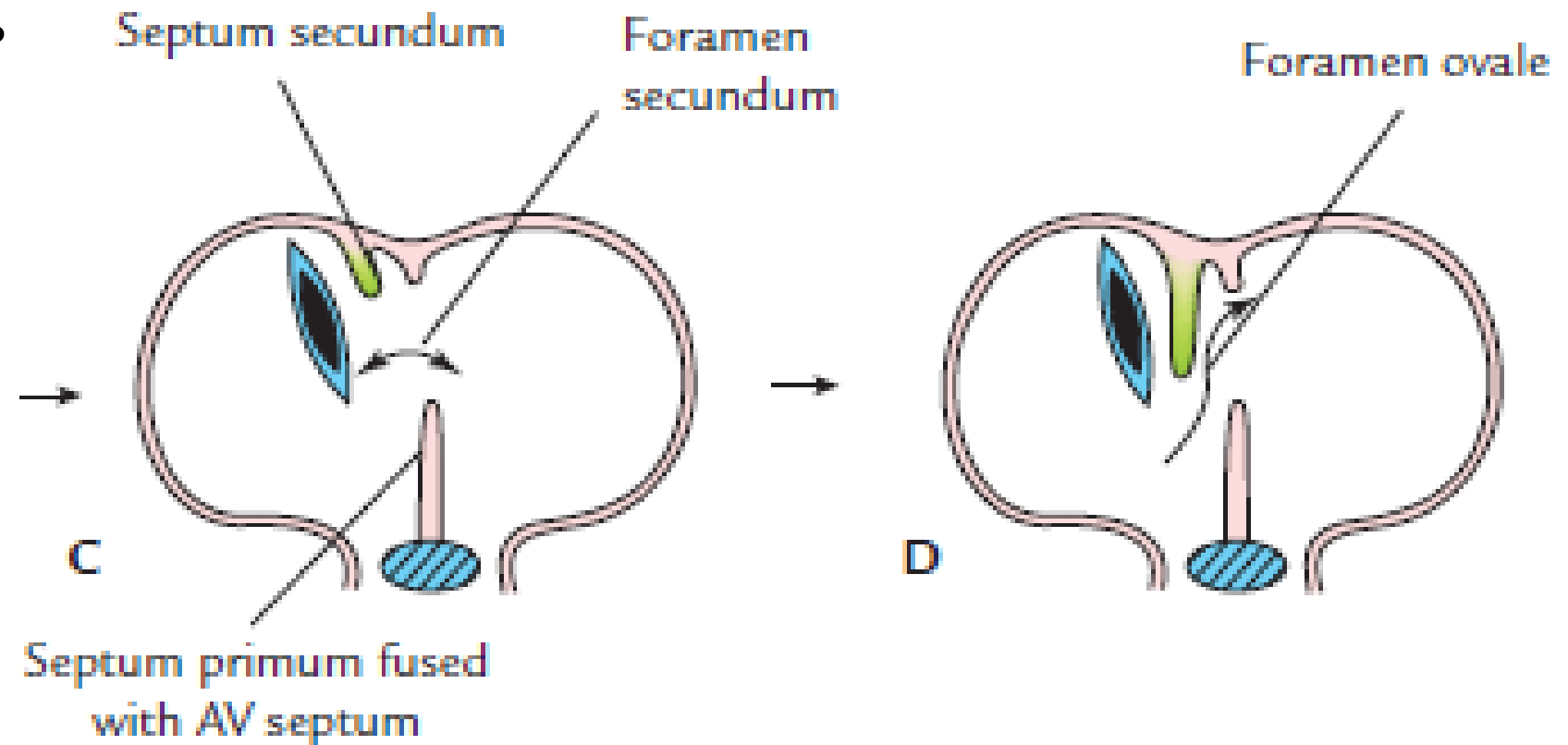


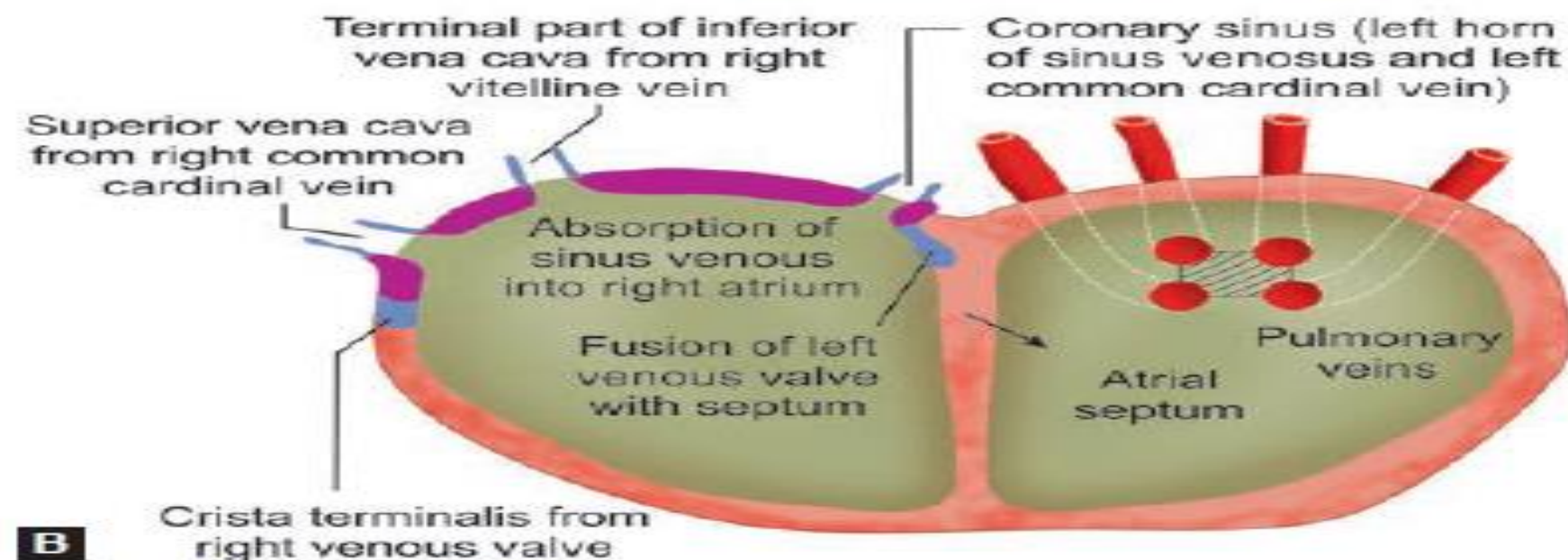
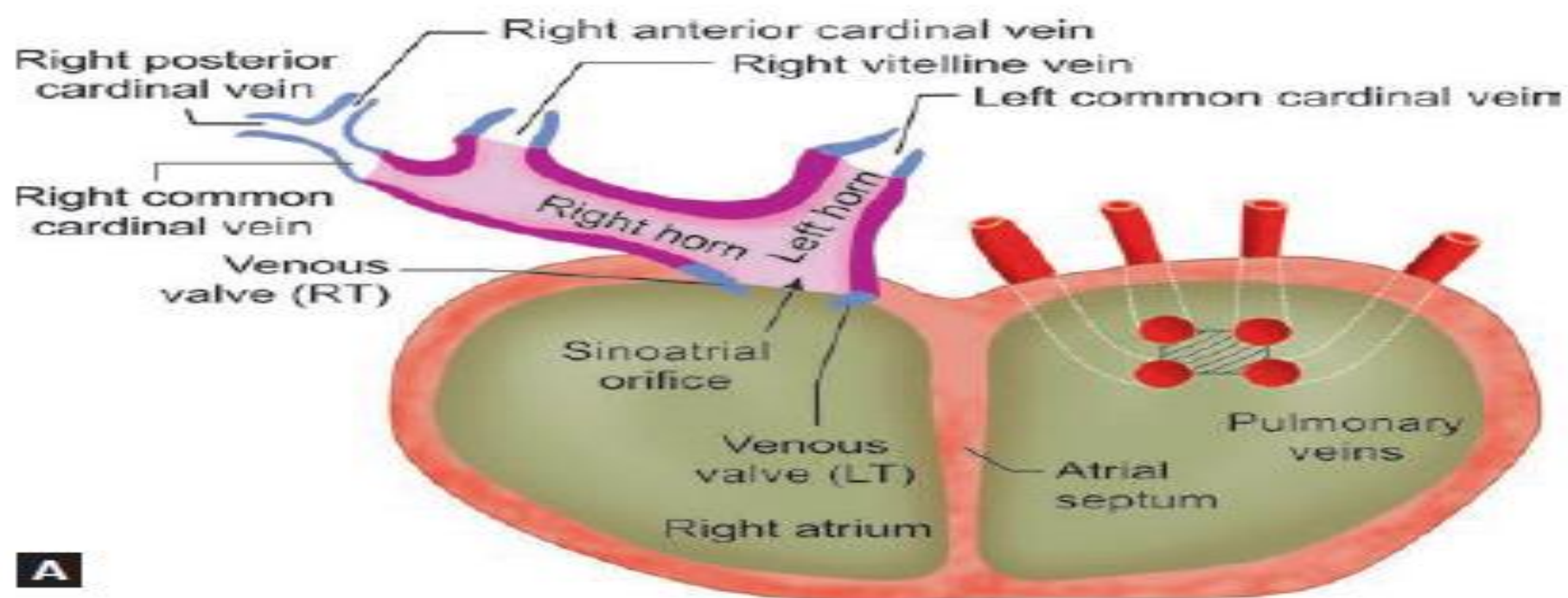
Fig. 6.5 Partitioning of the common atrium.

- ***septum secundum*** grows down from the roof of the atrial chamber, to the right of the septum primum
- grows, it comes to overlap the free upper edge of the septum primum
- left venous valve & cephalic attachment of septum primum get incorporated into the septum secundum
- the 2 septa overlap blood has to flow through the interval between the septa
- gap is the ***foramen ovale***



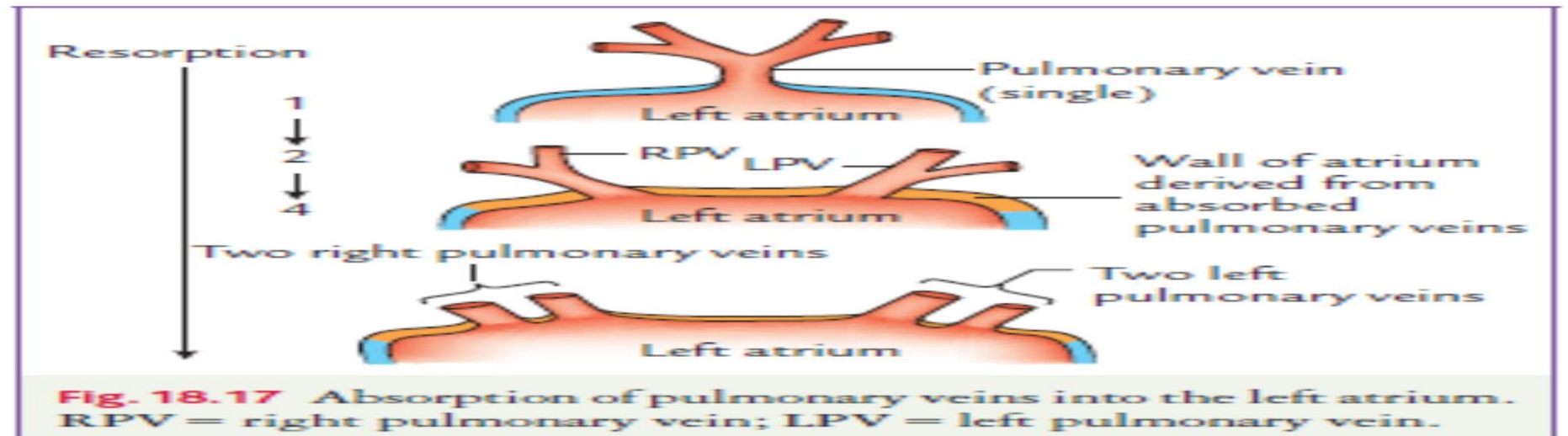
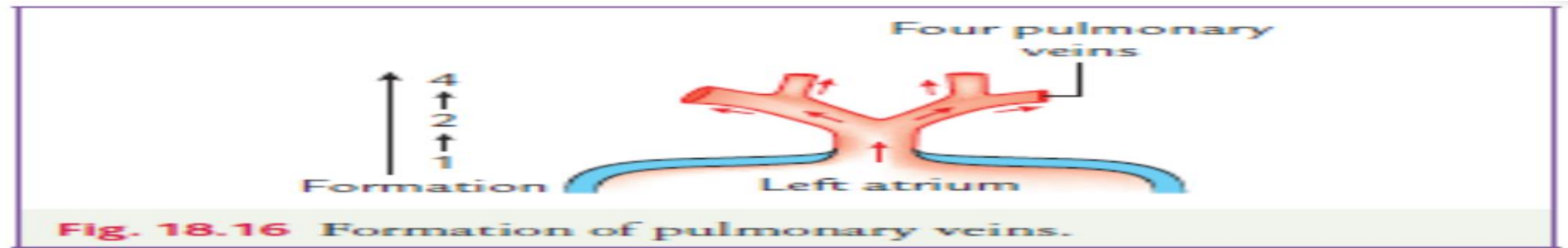
- foramen ovale is obliterated by fusion of the septum primum & septum secundum
- ***annulus ovalis*** represents the lower free edge of the septum secundum while ***fossa ovalis*** represents the septum primum in an adult

- ***Development of Right Atrium***
- main right atrium is derived from 3 sources
- 1. *Right half of the primitive atrium*: forms rough trabeculated part (atrium proper) in front of crista terminalis including right auricle
- 2. *Absorption of right horn of sinus venosus into the right half of primitive atrium*:
 - forms the smooth part (sinus venarum) behind crista terminalis develops from absorption of right horn and body of sinus venosus into the right atrium
- 3. *Absorption of right half of AV canal*:
 - crista terminalis lies at the junction of part of the right atrium derived from the sinus venosus (*sinus venarum*) & atrium proper



Figs 15.14A and B: Partitioned atrial chambers and sinoatrial orifice and pulmonary veins in partitioned atria

- ***Absorption of Pulmonary Veins into the Left Atrium***
- a single pulmonary vein opens into the left half of the primitive atrium
- vein divides into right & left branch each of which again bifurcates, to drain the corresponding lung bud
- parts of the pulmonary veins nearest to the left atrium are absorbed into the atrium, with the result that 4 separate veins, 2 from each side, come to open into it



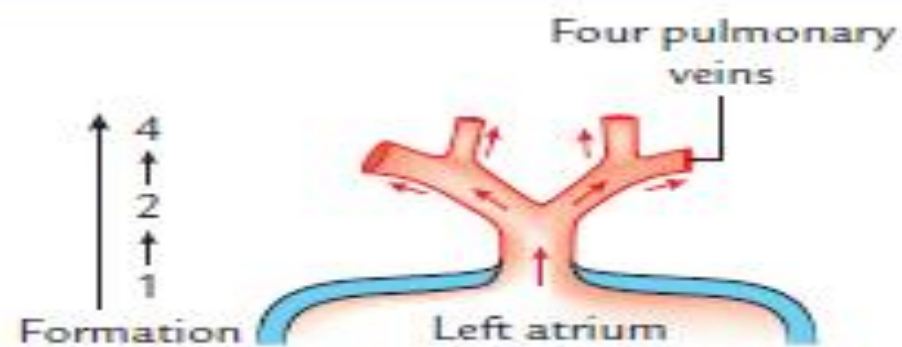


Fig. 18.16 Formation of pulmonary veins.

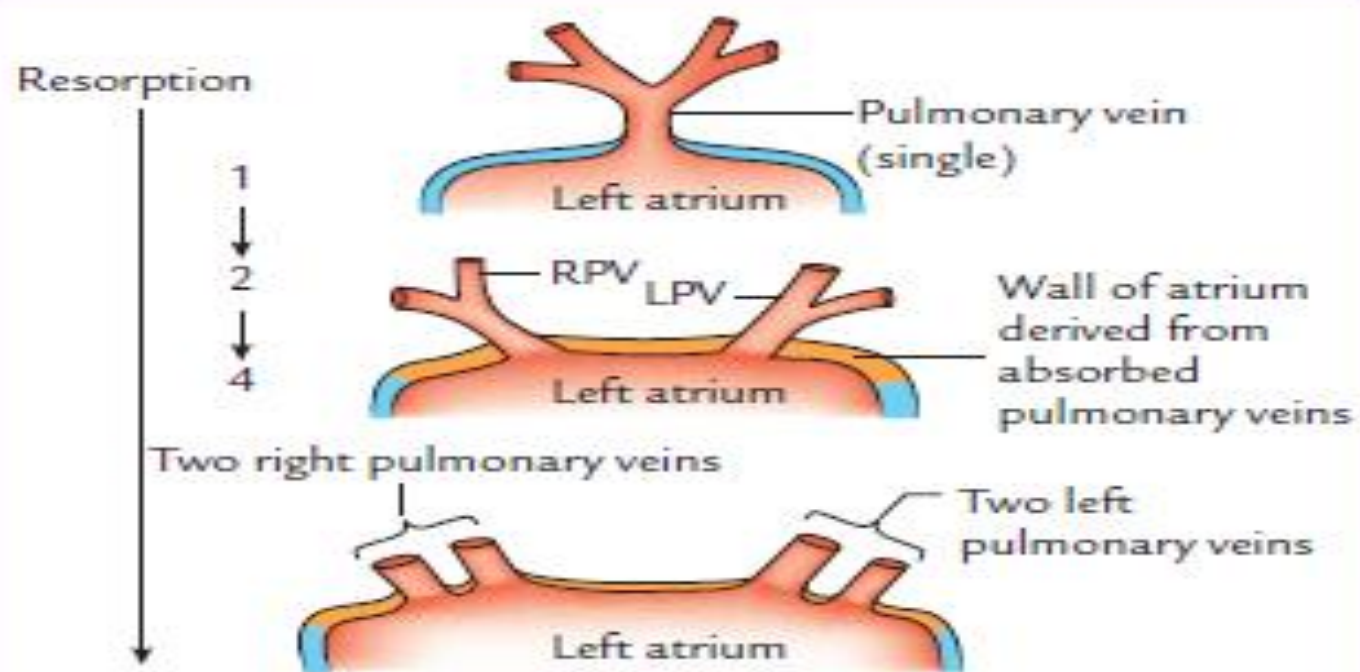


Fig. 18.17 Absorption of pulmonary veins into the left atrium.
RPV = right pulmonary vein; LPV = left pulmonary vein.

- ***Development of Left Atrium***
- left atrium is derived from the following 3 components
- 1. ***Left half of the primitive atrial chamber:***
- Contributes to rough anterior part & left auricle.
- 2. ***Left half of the AV canal:***
- forms the most anterior smooth part
- 3. ***Absorbed proximal parts of the pulmonary veins:***
- forms posterior smooth part between the openings of pulmonary veins that form the anterior boundary for oblique sinus of pericardium.

- **Development of Ventricles**
- right & left ventricles formed by partitioning of primitive ventricle & incorporation of bulbus cordis
- ***Bulbus Cordis***
- is the cranial most part (arterial end) of the developing heart tube
- It is divisible into 3 parts:-
 1. **proximal one-third** is dilated & does not have any special name
 2. **middle one-third** is the **conus**
 3. **distal one-third** is the **truncus arteriosus**

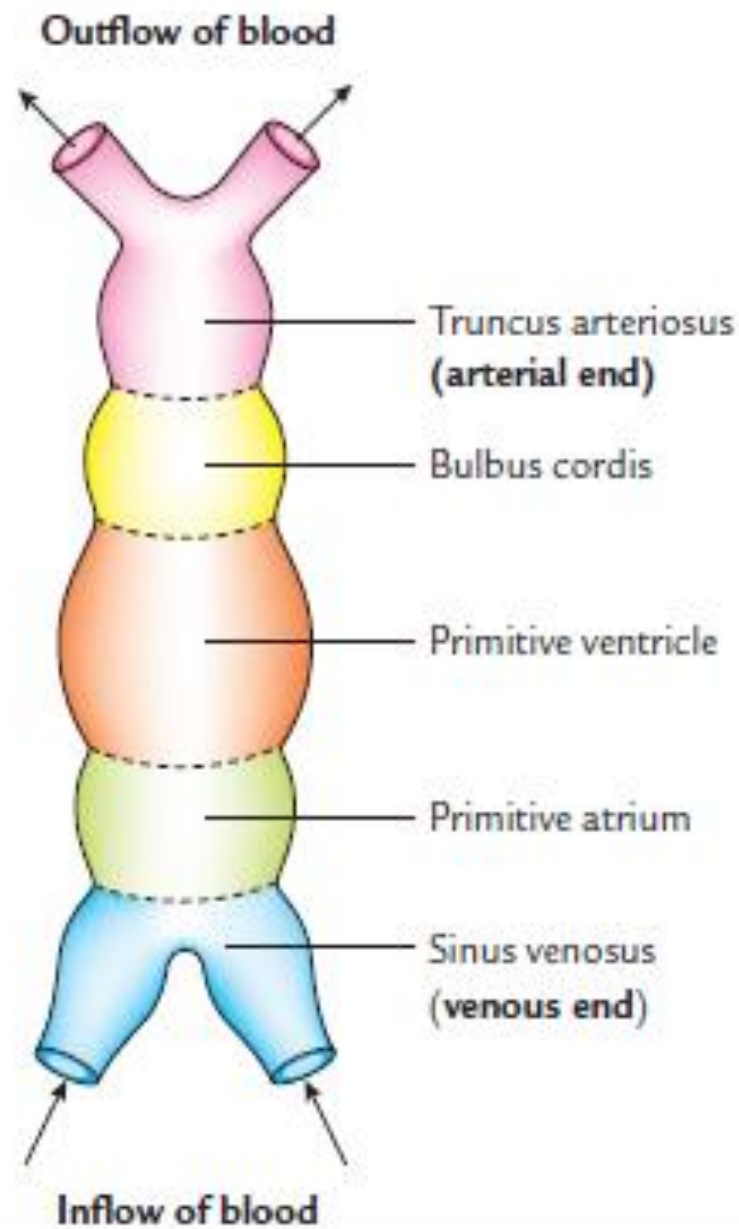
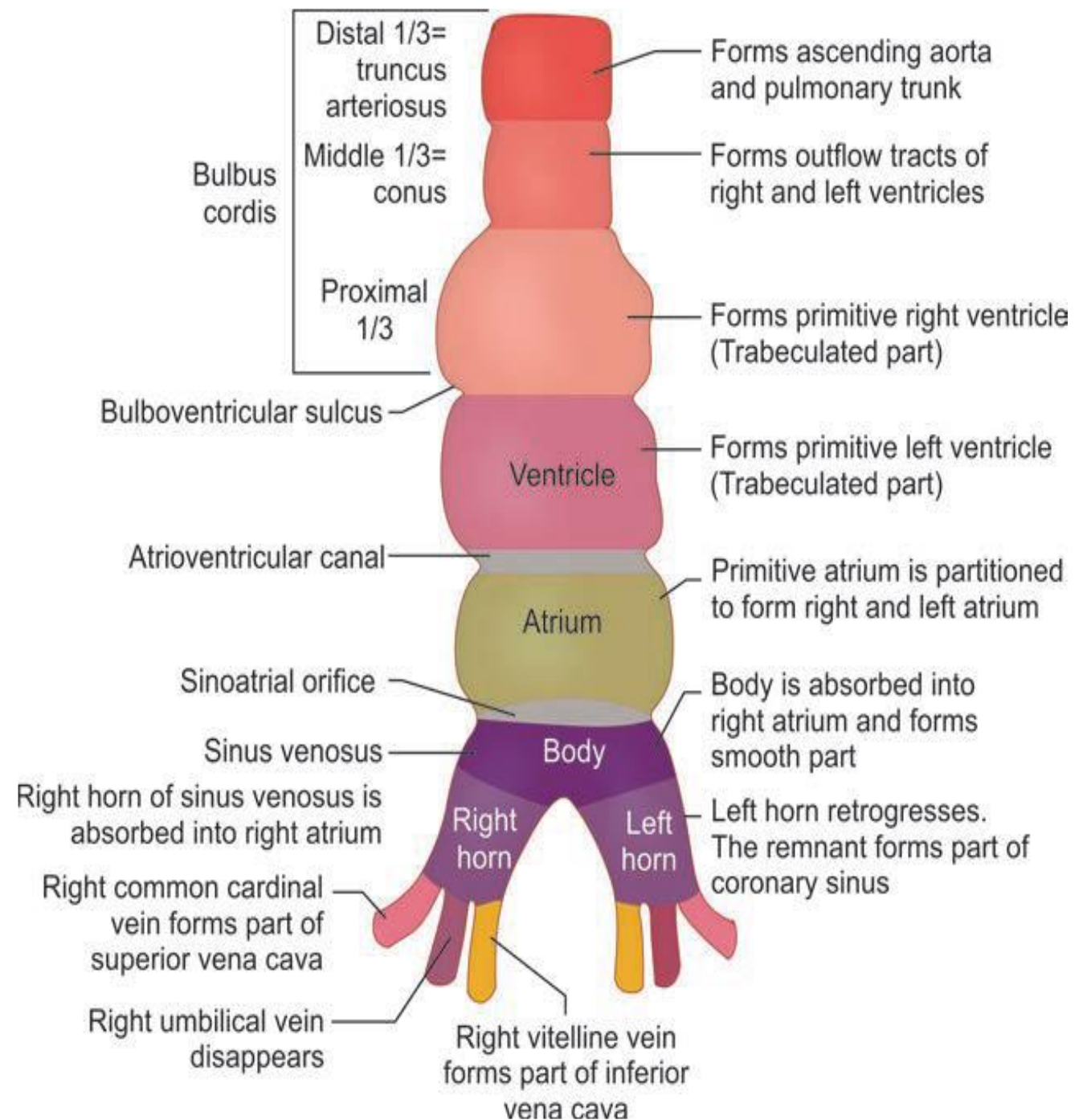


Fig. 18.3 Five embryonic dilatations (subdivisions) of the heart tube.



- **proximal one-third** merges with the cavity of the primitive ventricle & forms the ***bulboventricular chamber***
- takes part in forms the trabeculated part of the right ventricle
- **middle part (*conus cordis*)** forms the outflow part of both the ventricles
- 2 septa are formed in relation to the walls of conus
- ***proximal & distal bulbar septa***
- proximal bulbar septum contributes to the formation of ***interventricular septum***
- Distal bulbar septum separates the conus into ***aortic vestibule & conus arteriosus/infundibulum***

- distal part (*truncus arteriosus*) undergoes division by spiral *aortopulmonary septum* into *ascending aorta* & *pulmonary trunk*
- **spiral septum** is formed by union of **right superior & left inferior *truncus swellings* or *cushions***
- truncus arteriosus is continuous distally with the aortic sac
- aortic sac is continuous with right and left pharyngeal arch arteries

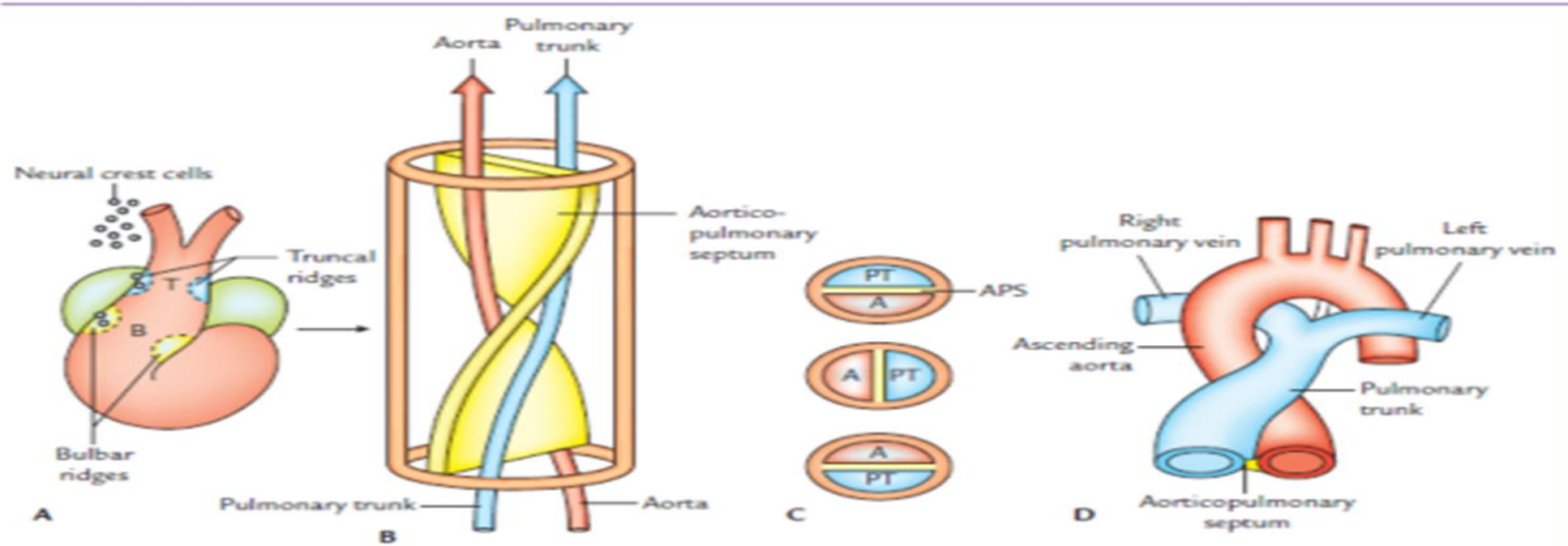


Fig. 18.23 Formation of aorticopulmonary septum. A. Partitioning of truncus arteriosus and bulbus cordis involves migration of neural crest cells from truncal and bulbar ridges. B. Aorticopulmonary septum develops in a spiral fashion. C. Sections through the newly formed aorta (A) and pulmonary trunk (PT) showing the position of aorticopulmonary septum in upper, middle, and lower parts. D. Adult gross anatomical relationship between ascending aorta and pulmonary trunk. T = truncus arteriosus. B = bulbus cordis; PT = pulmonary trunk; APS = aorticopulmonary septum.

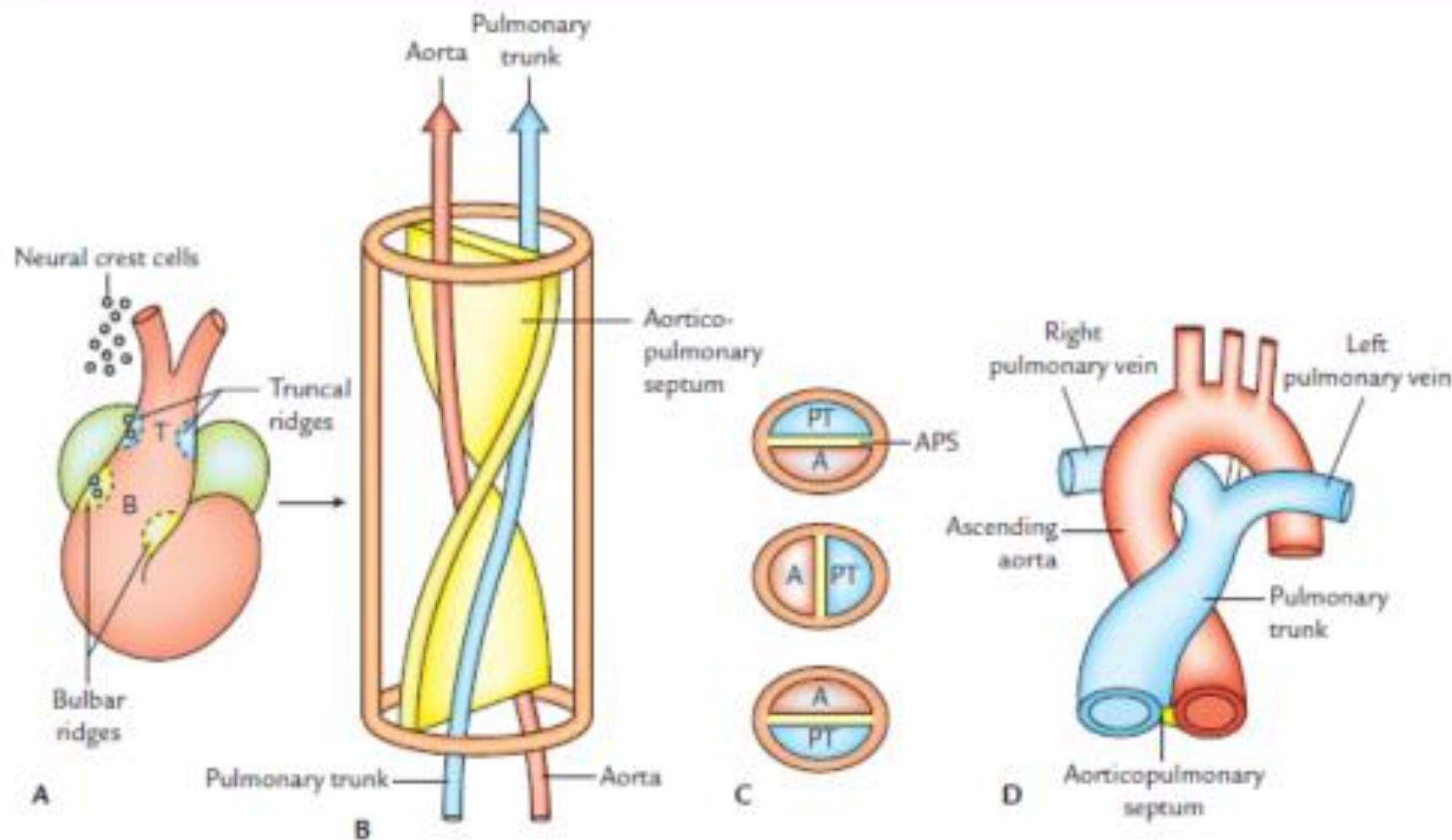


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- ***Formation of Interventricular Septum***
- ventricular cavity formed after the conus & proximal one-third of bulbus cordis has merged into the primitive ventricle
- Has to subdivided into right & left , each half has to communicate with the corresponding atrium
- right ventricle opens into the pulmonary trunk & the left ventricle into the aorta
- subdivision takes place by the formation of interventricular septum

- interventricular septum consists of 3 parts that develop from different sources
- (1) *muscular*, (2) *bulbar* and (3) *membranous* parts
- ***Muscular part :***
- A septum, the *interventricular septum*, grows upward *from the floor of the bulboventricular cavity* and divides the lower dilated part of this cavity into right and left halves
- meets the fused AV cushions (*septum intermedium*) and partially fuses with them

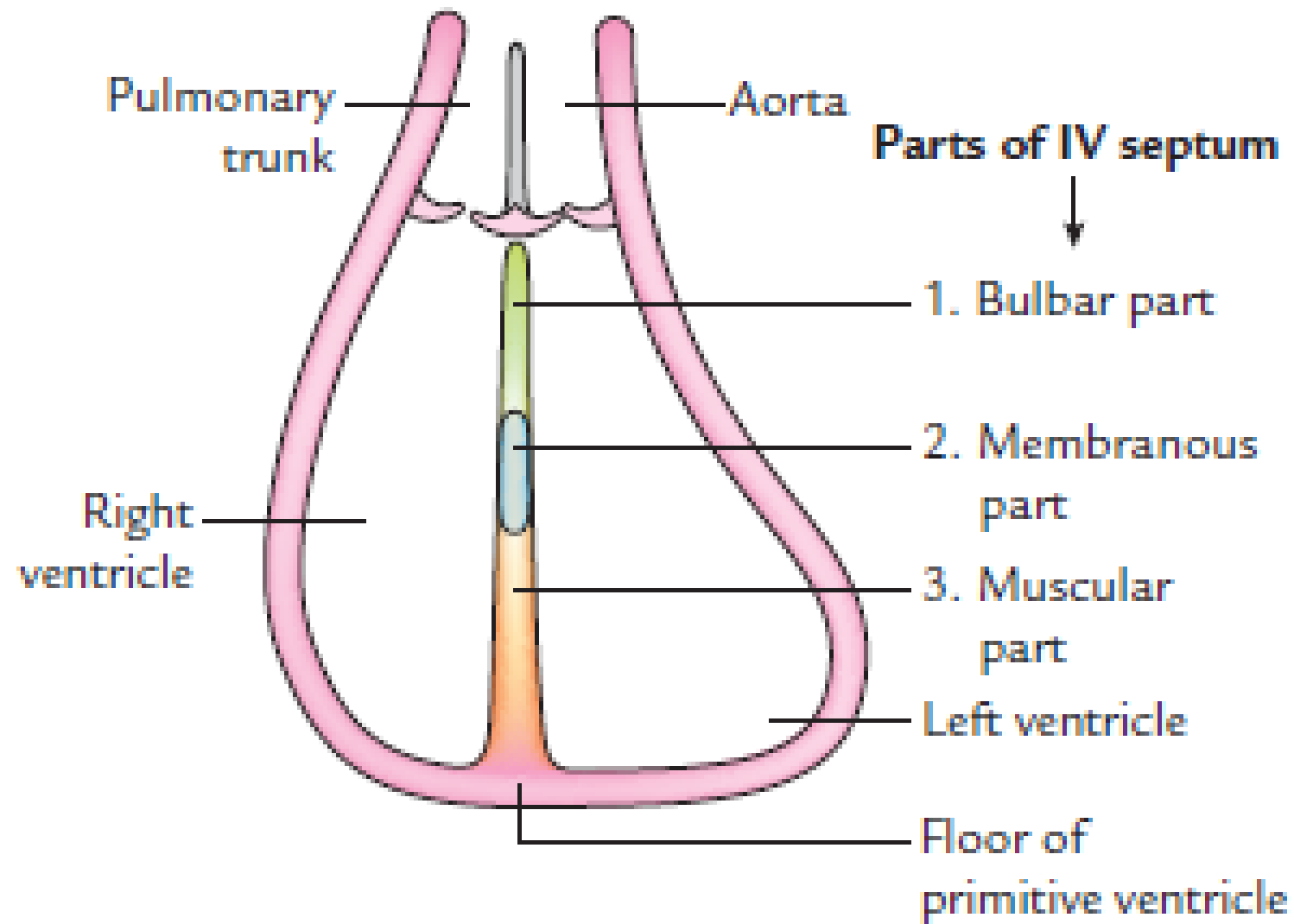
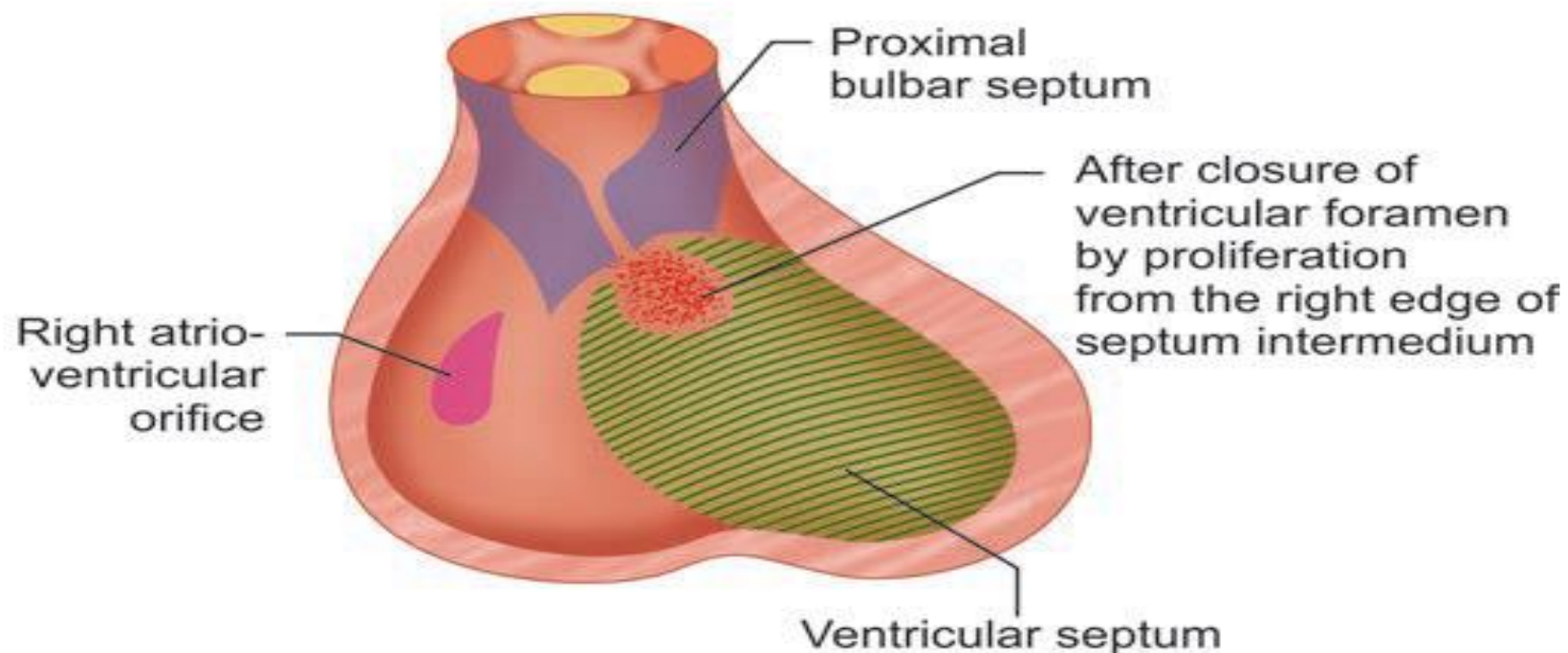
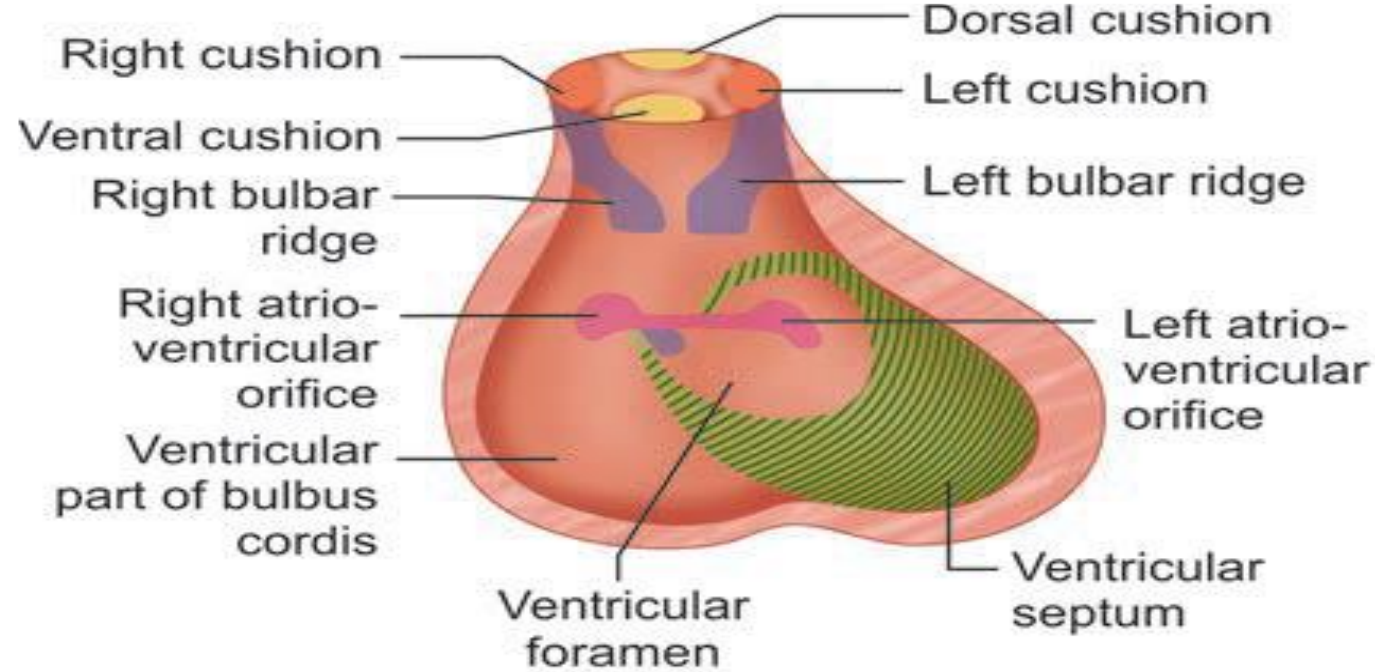
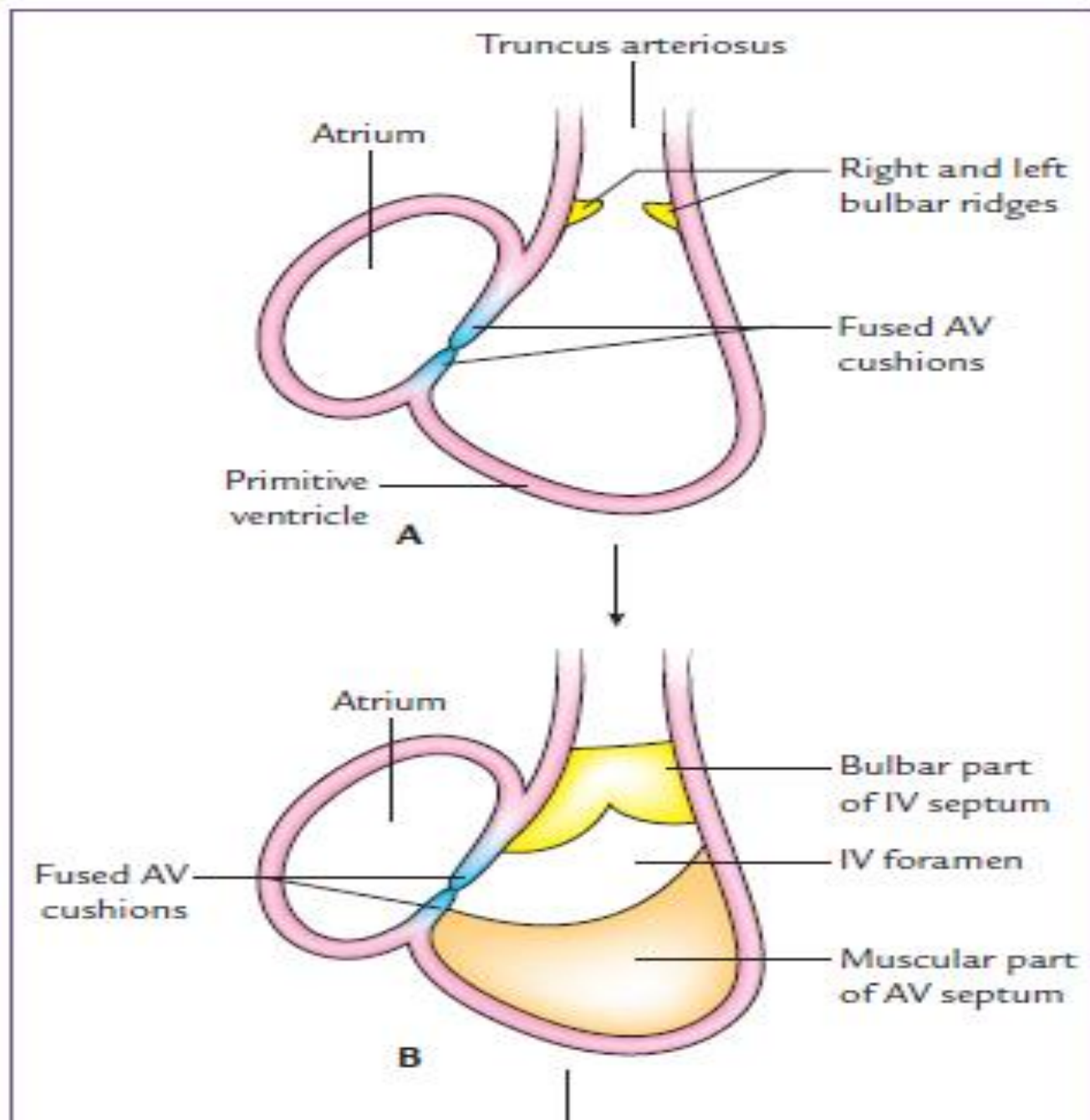


Fig. 18.19 Components of the interventricular (IV) septum.

- cephalic margin of septum is free, concave and twisted
- presents a *dorsal* and a *ventral horn*
- dorsal horn fuses with right edge of dorsal AV cushion and the ventral horn fuses with the left edge of ventral AV cushion
- **interventricular foramen** appears between the 2 ventricles at the upper margin of interventricular septum
- Closure of interventricular foramen is facilitated by **septum intermedium** and **proximal bulbar septum**



- **Bulbar part develops from right and left bulbar ridges:**
- ***right and left bulbar ridges***, arise in the wall of the bulboventricular cavity (in the conus)
- right ridge arises from the dorsal and right wall and is cephalic to right AV orifice
- left ridge arises from the ventral and left wall
- Ridges grow toward each other and fuse to form a ***bulbar septum***
- right ridge is in line with the dorsal horn and the left ridge with the ventral horn of muscular part of AV septum
- Fusion of right and left bulbar ridges forms the **proximal bulbar septum**



- bulbar septum grows downward towards muscular part of interventricular septum does not reach it
- a gap is still left between the two
- ***Membranous part:***
- gap between upper edge of the interventricular septum, and lower edge of the bulbar septum filled by proliferation of tissue from
- the **right side of the AV cushions** and the **right and left bulbar ridges**
- *membranous part of the interventricular septum* is divisible into :-
 - ❖ an anterior part, which separates the right and left ventricles
 - ❖ posterior part which separates the left ventricle from the right atrium

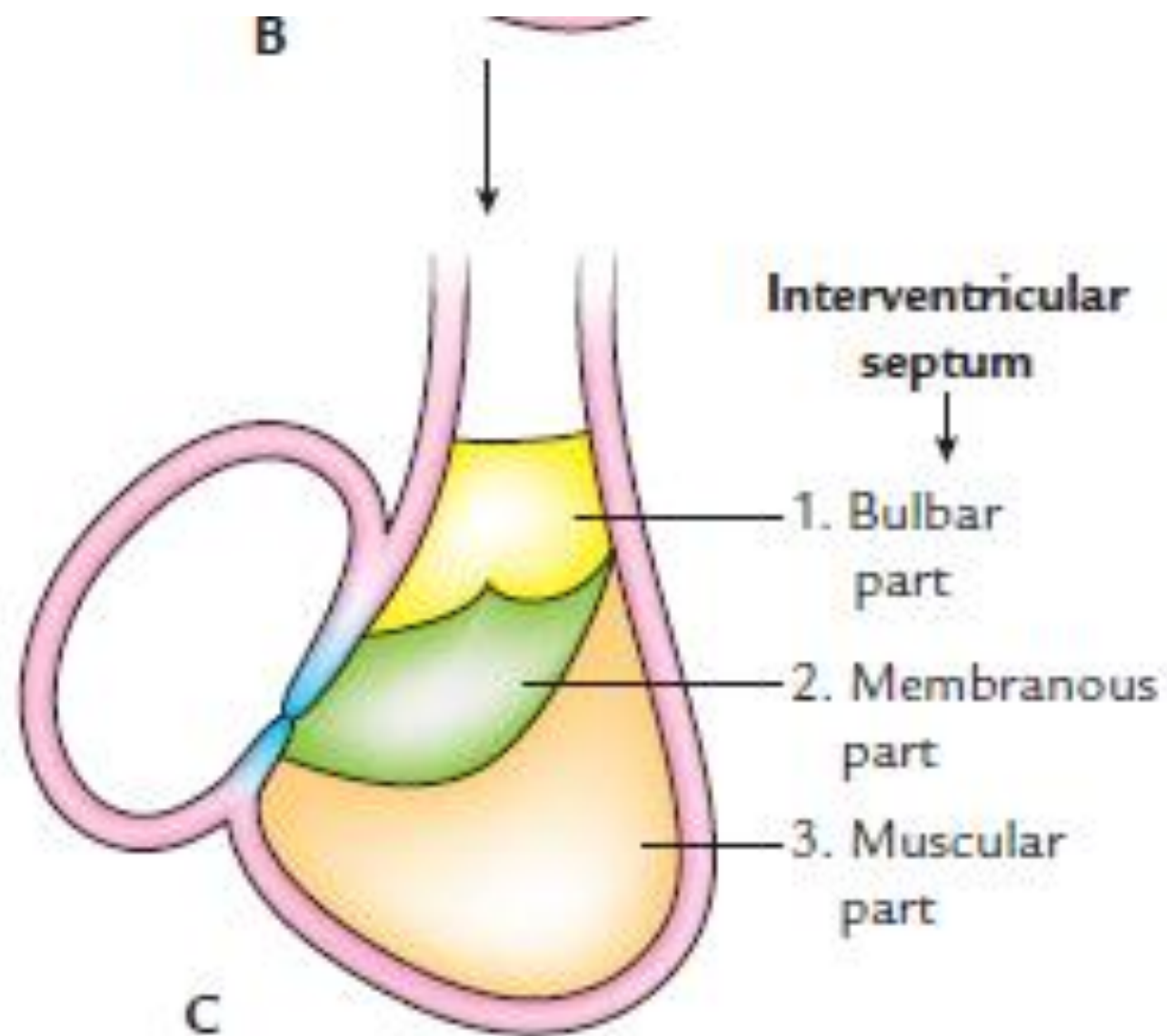


Fig. 18.20 Development of interventricular septum.

- **Clinical Correlation**

- 1. ***Dextrocardia***

- all the chambers of the heart and associated blood vessels are reversed as a mirror image, i.e., all the structures that normally lie on to the right side are on to the left
- occurs if heart tube bends to the left instead of the right, the heart is displaced to the right, and there is transposition
- may be associated with **situs inversus** (transposition of abdominal viscera)
- **dextrocardia is the most common positional anomaly of the heart**

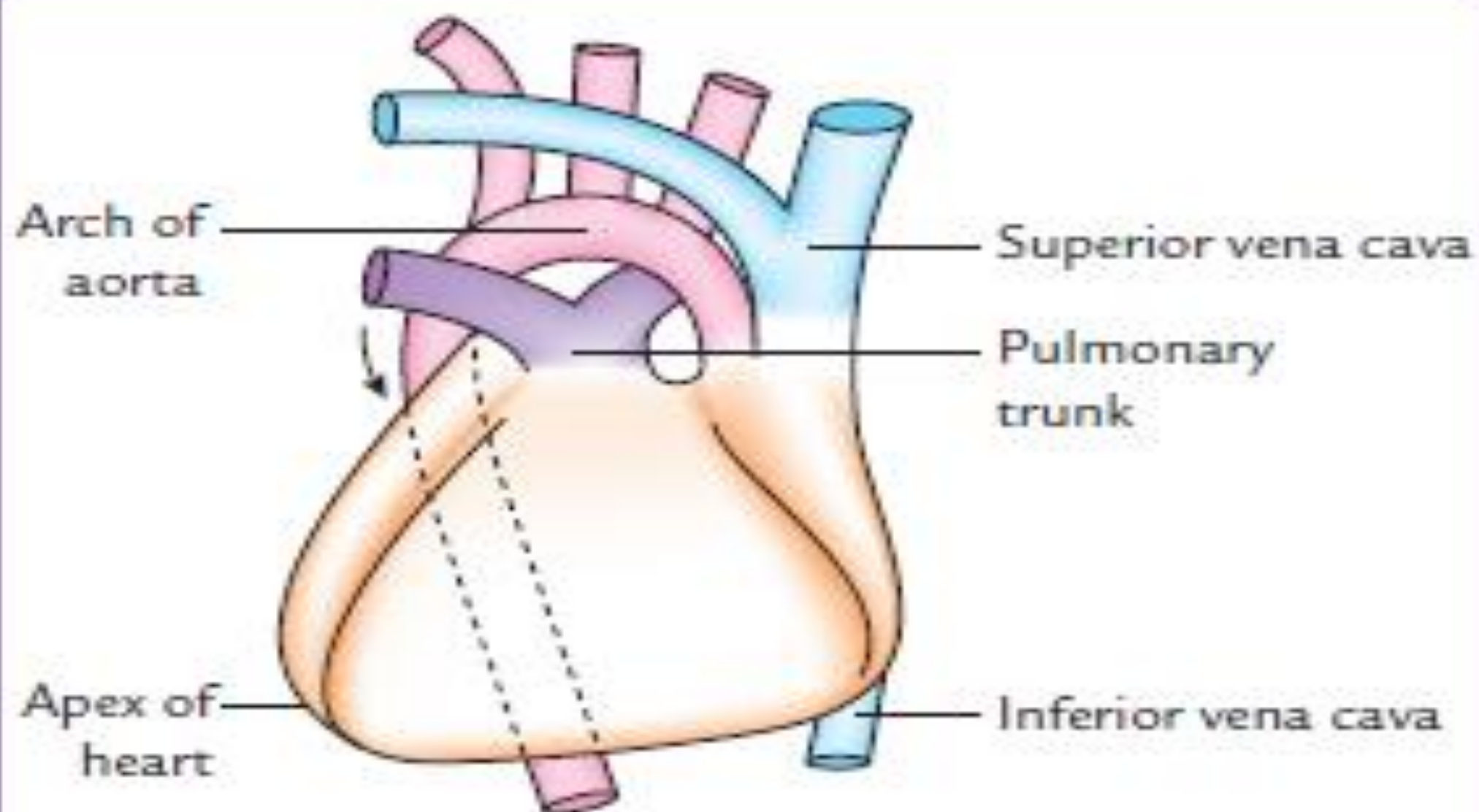


Fig. 18.9 Dextrocardia.

- **2. *Ectopia cordis***

- rare condition in which the heart lies exposed on the surface of the thorax
- occurs due to nonunion of two sternal plates of developing sternum
- Death occurs in most of the cases during first few days of the birth

- **3. *Atrial septal defects***

- There is a defect in interatrial septum, which results in the communication between right and left atrium
- There is a mixing of oxygenated and deoxygenated blood thereby giving rise to symptoms like cyanosis, breathlessness, etc

Heart exposed to the exterior



Fig. 18.10 A newborn baby with ectopia cordis.

- **1. Septum primum defect:**
- The septum primum fails to reach the septum intermedium
- As a result, *persistent foramen primum* persists
- **2. Septum secundum defect:**
- occurs if the septum secundum does not grow downward adequately to overlap the septum primum
- or the septum primum may undergo *excessive resorption*, because of which septum secundum fails to overlap the septum primum.
- As a result, foramen secundum remains large and unclosed—*ostium secundum defect*
- This results in **patent foramen ovale**
- It is the most significant atrial septal defect

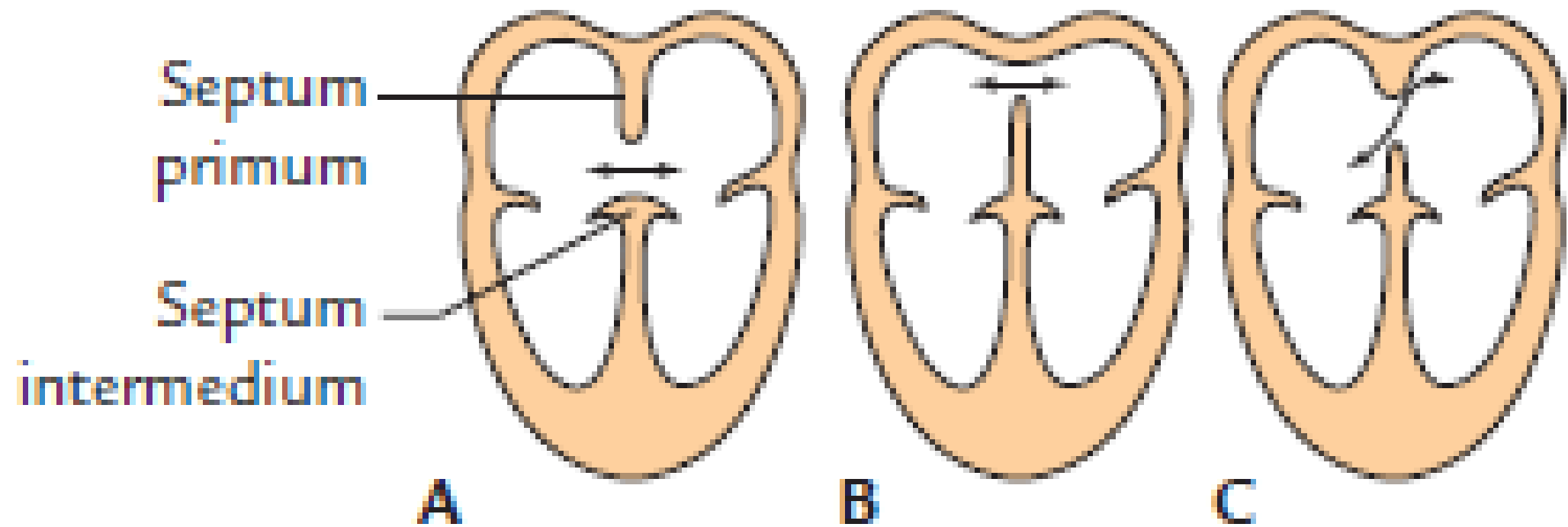


Fig. 18.18 Atrial septal defects.

- **3. Persistent foramen ovale [probe patency of foramen ovale]**
- oblique passage between septum primum and septum secundum is closed functionally but remains patent anatomically so that a probe can be passed through it
- The ASD is most common congenital anomaly of the atria.
- If the ASD is small, the clinical symptoms may be delayed up to the 30 years of age

- **4. Premature (prenatal) closure of foramen ovale:**
 - closure of foramen ovale occurs during prenatal life
 - This results in the massive hypertrophy of the right side of the heart and underdevelopment of the left side of the heart
- **5. Cor triloculare-biventricular/common atrium:**
 - a complete absence of the interatrial septum
 - results in three-chambered heart
 - most serious congenital anomaly of the atria and is always associated with other defects of the heart

- 1. ***Ventricular septal defect (VSD)***
- most common congenital anomaly of the heart
- defect is most common in the membranous part of the IV septum
- caused by failure of fusion of right and left bulbar ridges with the AV cushions, which form the membranous part of the IV septum
- occurs a defect in the IV septum that results in communication between right and left ventricles
- **Blood flows from the left to the right ventricle**
- a result of left-to-right shunting of blood, the output from the left ventricle is reduced

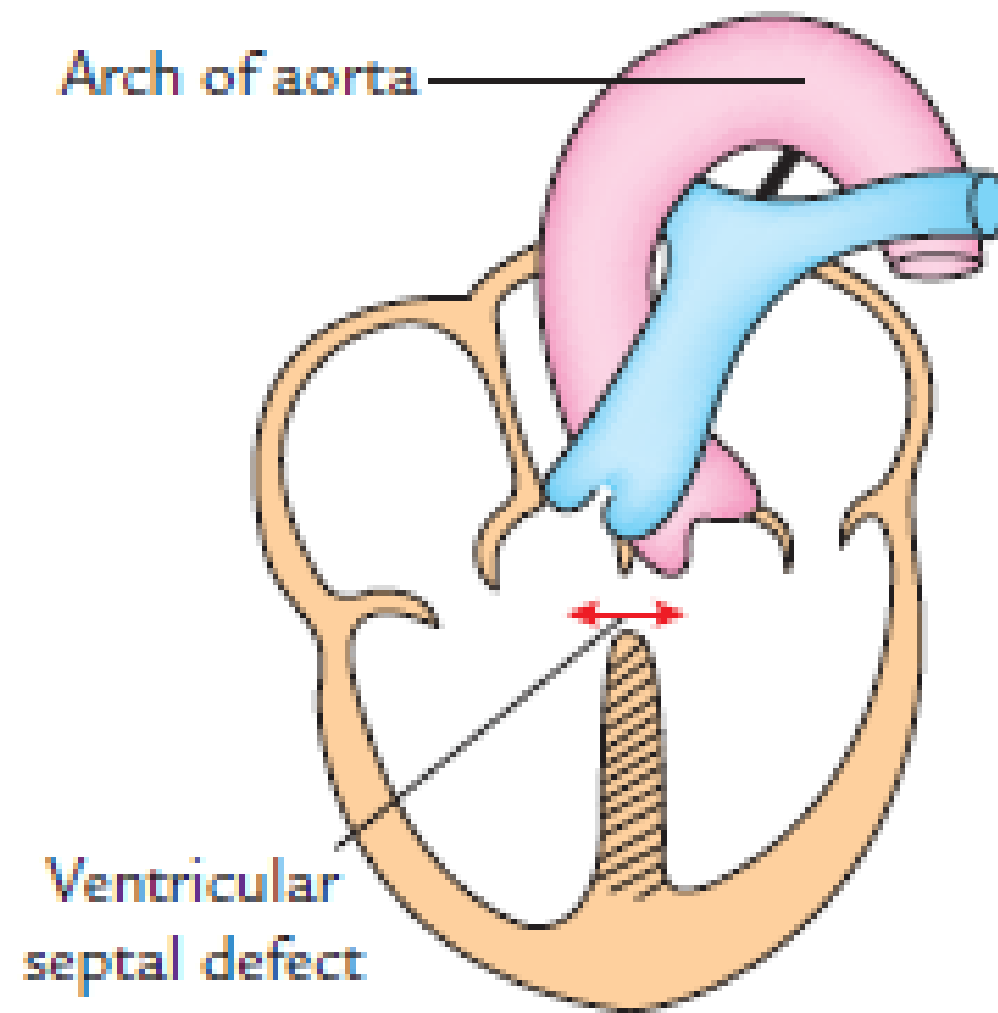


Fig. 18.21 Ventricular septal defect.

- patient complaints of excessive fatigue on exertion
- Rarely, the defect may be present in muscular part of the IV septum
- **2. Fallot's tetralogy [tetralogy of Fallot]**
- condition occurs due to combinations of four cardiac anomalies
- (a) **pulmonary stenosis,**
- (b) **overriding aorta** (i.e. displacement of aortic orifice to the right to override the IV septum)
- (c) **IV septal defect**
- (d) **hypertrophy of the right ventricle**
- resultant right-to-left shunting of blood leads to cyanosis
- *The Fallot's tetralogy is the commonest congenital cyanotic heart disease*

1. Stenosis of pulmonary trunk
2. Ventricular septal defect
3. Overriding of aorta
4. Right ventricular hypertrophy

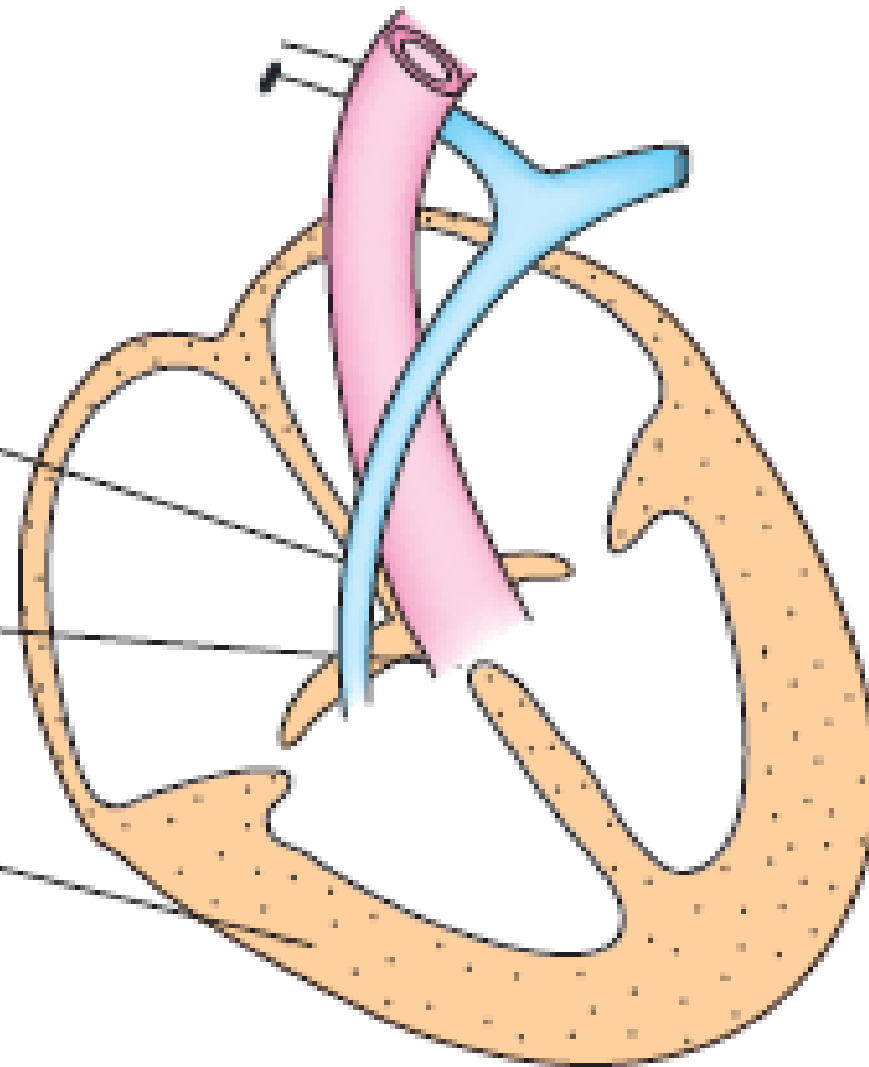


Fig. 18.22 Fallot's tetralogy.

- ***Anomalies of truncus arteriosus***
- **1. Persistent truncus arteriosus:**
 - In this anomaly spiral septum (aortopulmonary septum) fails to develop
 - Pulmonary arteries and aorta arise from a common vessel
- **2. Defect in development of spiral septum**
 - results in communication between ascending aorta and pulmonary trunk
- **3. Transposition of great vessels**
 - In this anomaly ascending aorta arises from the right ventricle and pulmonary trunk arises from the left ventricle.
 - It occurs due to nonspiral development of aortopulmonary septum

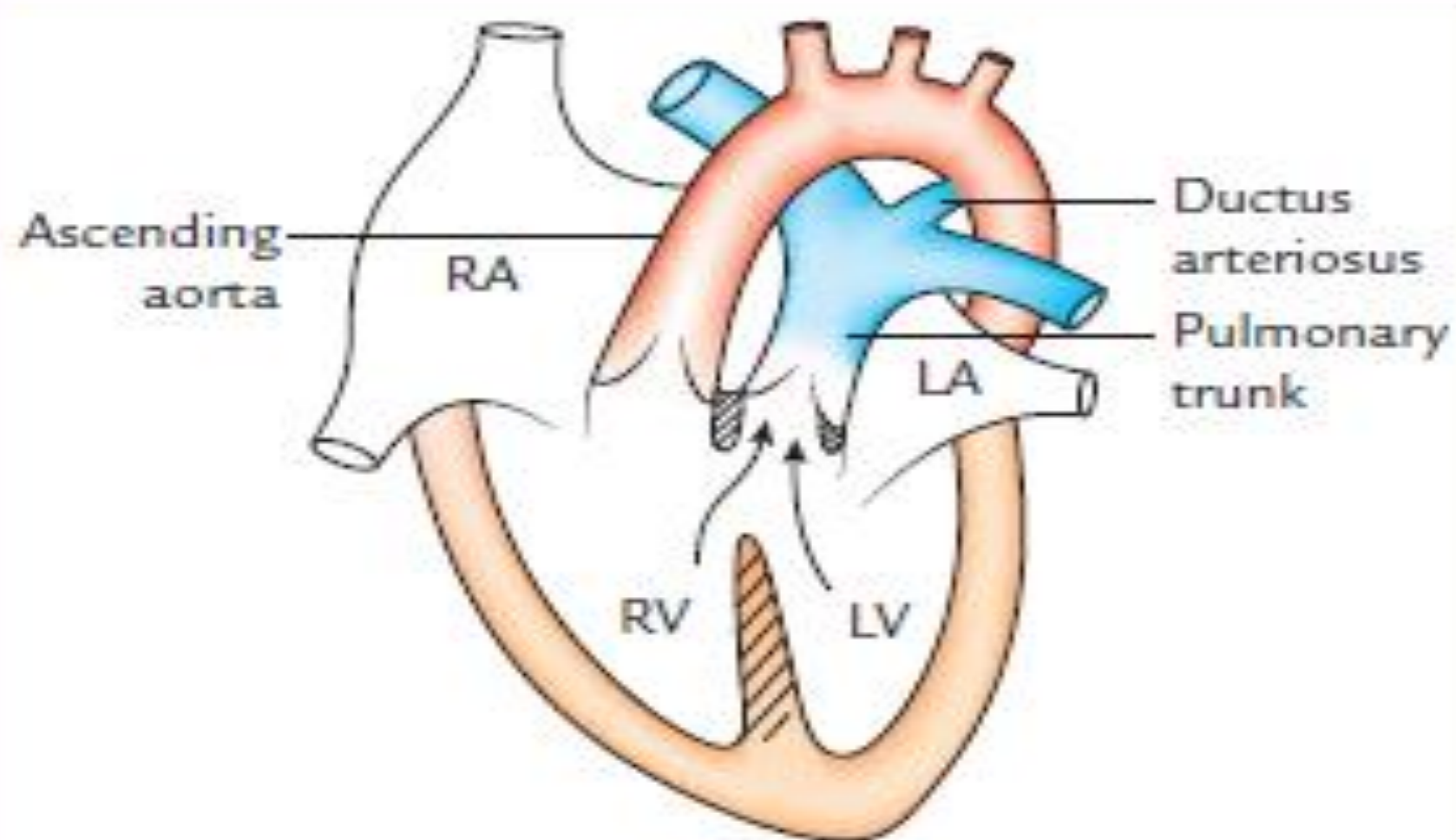


Fig. 18.24 Transposition of great vessels. RA = right atrium; LA = left atrium; RV = right ventricle; LV = left ventricle.

- ***Tricuspid atresia***
- there is complete agenesis of tricuspid valves
- It is caused by an insufficient amount of tissue in AV cushion for the formation of tricuspid valve
- associated with
 - (a) patent foramen ovale
 - (b) VSD
 - (c) underdeveloped right ventricle
 - (d) hypertrophy of the right ventricle

- ***Atresia or stenosis***
- Any of the orifices of the heart may have too narrow an opening (stenosis), or none at all (atresia)
- aortic and pulmonary passages may also show supravalvular, or subvalvular, stenosis
- Alternatively, the openings may be too large as a result of which the valves become incompetent
- pulmonary stenosis, the foramen ovale and the ductus arteriosus remain patent
- In aortic stenosis also, the ductus arteriosus is patent and blood flows into the aorta through it

- <https://youtu.be/Wv5APMkXlyg>
- <https://youtu.be/B9VNg3zsZ2c>
- <https://youtu.be/t-YEUTLbkMI>
- <https://youtu.be/Ob7oevkGCAw>
- Assignment: Study the development of the heart valves and conducting system and fetal circulation